

A Unified Framework for Adsorption Isotherm Analysis in Water Quality Studies: Descriptors, Model Selection, and Reporting Guidelines

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How to read this SI

This SI is written as a practical route from raw isotherm data to reportable descriptors. For each model, ask three questions: **(1)** does it have a real saturation capacity $Q_{\max}$? **(2)** does it provide a comparable affinity K_{aff}^* ? **(3)** what heterogeneity descriptor should be reported?

64

Nomenclature used for reporting

The final reporting descriptors are

$$\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}.$$

Here $\Delta G_{\text{ads}}^0 = -RT \ln K_{\text{aff}}^*$ is the standard Gibbs free energy associated with the declared standard state. The reported positive affinity-energy descriptor is $E_{\text{ads}} \equiv E_{\text{aff}} = -\Delta G_{\text{ads}}^0 = RT \ln K_{\text{aff}}^*$. Larger E_{ads} therefore means stronger standard-state affinity. The standard-state concentration may appear as C° or C^0 ; both mean the declared concentration standard used to make equilibrium constants dimensionless. The symbol σ_E denotes the apparent mathematical width of an energy distribution; σ_H is the reported intrinsic heterogeneity width.

65

Warning

Do **not** report $Q_{\max}$ from the Freundlich equation. The Freundlich model has no saturation plateau; any capacity derived from it depends on the chosen concentration window and is physically meaningless. **FREUNDLICH**

66

S-1 Isotherm model equations and limits

This section collects the full equations and limiting behaviors used in Section 2 of the main manuscript. The availability summary is given in Table 1 of the main manuscript.

Reader rule

For every isotherm, separate the fitted parameters from the reported descriptors. Fitted parameters belong to one mathematical model; descriptors are the quantities used to compare materials across different models.

S-1.1 Henry isotherm

The Henry isotherm is the low-coverage limit of many models:

$$q_e = K_H C_e. \quad (\text{S-1})$$

It has no saturation capacity $Q_{\max}$ and therefore no model-harmonized $K_{\text{aff}} = K_H/Q_{\max}$.

The Henry constant K_H is defined by the initial slope.

Low-concentration limit. The Henry model is itself the $C_e \rightarrow 0$ limit:

$$\lim_{C_e \rightarrow 0} \frac{q_e}{C_e} = K_H. \quad (\text{S-2})$$

High-concentration limit. The model is linear and **does not saturate**, so $q_e \rightarrow \infty$ as

$C_e \rightarrow \infty$.

Take-home message

Henry's law is useful for the initial slope, but it is not a full capacity model. Use K_H to describe dilute uptake; do not use the Henry equation to report $Q_{\max}$.

79 S-1.2 Langmuir isotherm

80 The Langmuir isotherm is written as

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e}, \quad (\text{S-3})$$

81 where q_e is the equilibrium adsorbed amount, C_e is the equilibrium concentration in the
82 fluid phase, $Q_{\max}$ is the maximum adsorption capacity, and K_L is the Langmuir equilibrium
83 constant.

84 S-1.2.1 High concentration limit, $Q_{\max}$: $C_e \rightarrow \infty$

85 From Eq. (S-3),

$$\lim_{C_e \rightarrow \infty} q_e = \lim_{C_e \rightarrow \infty} \frac{Q_{\max} K_L C_e}{1 + K_L C_e} = Q_{\max}. \quad (\text{S-4})$$

86 Therefore,

Capacity-limit result

Langmuir has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-5})$$

This makes Langmuir the clean reference model for reporting $Q_{\max}$ and for defining the
homogeneous limit $\sigma_H = 0$.

88 S-1.2.2 Low concentration limit: Henry constant

89 For small C_e with $K_L C_e \ll 1$,

$$q_e \approx Q_{\max} K_L C_e. \quad (\text{S-6})$$

90 The Henry law form is

$$q_e \approx K_H C_e, \quad (\text{S-7})$$

91 so the Henry constant is the initial slope:

$$K_H = \left. \frac{dq_e}{dC_e} \right|_{C_e \rightarrow 0} = Q_{\max} K_L. \quad (\text{S-8})$$

92 Thus,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = Q_{\max} K_L. \quad (\text{S-9})$$

93

94 **S-1.2.3 Adsorption free energy**

95 Introduce a standard concentration C^0 and define

$$K_L^* = K_L C^0. \quad (\text{S-10})$$

96 The standard Gibbs free energy of adsorption is

$$\Delta G_{\text{ads}}^0 = -RT \ln (K_L C^0), \quad (\text{S-11})$$

97 and the corresponding affinity energy is

$$E_{\text{aff}} = -\Delta G_{\text{ads}}^0 = RT \ln (K_L C^0). \quad (\text{S-12})$$

98 Equivalently,

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$\Delta G_{\text{ads}}^0 = -RT \ln (K_L C^0), \quad E_{\text{aff}} = RT \ln (K_L C^0). \quad (\text{S-13})$$

Take-home message

Langmuir is the reference homogeneous model. It gives a finite Q_{max} , a finite Henry constant, and $\sigma_H = 0$. Any heterogeneous model that becomes Langmuir in a limiting case should reduce smoothly to these three values.

S-1.3 Double Langmuir isotherm

The dual-site (double) Langmuir isotherm is

$$q_e = \frac{Q_{\text{max},1} K_1 C_e}{1 + K_1 C_e} + \frac{Q_{\text{max},2} K_2 C_e}{1 + K_2 C_e}, \quad (\text{S-14})$$

where $Q_{\text{max},i}$ and K_i are the site capacities and constants.

S-1.3.1 High concentration limit: $C_e \rightarrow \infty$

For each site,

$$\lim_{C_e \rightarrow \infty} \frac{Q_{\text{max},i} K_i C_e}{1 + K_i C_e} = Q_{\text{max},i}, \quad (\text{S-15})$$

so the total saturation capacity is finite.

Capacity-limit result

Double Langmuir has a finite total saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max,1} + Q_{\max,2}. \quad (\text{S-16})$$

Report the sum as the total capacity, but keep $Q_{\max,1}$, $Q_{\max,2}$, K_1 , and K_2 when the goal is to interpret two distinct site families.

S-1.3.2 Low concentration limit: global Henry constant

For $K_i C_e \ll 1$,

$$q_e \approx Q_{\max,1} K_1 C_e + Q_{\max,2} K_2 C_e, \quad (\text{S-17})$$

which has the Henry form (Eq. (S-7)) with

$$K_H^{(\text{tot})} = Q_{\max,1} K_1 + Q_{\max,2} K_2. \quad (\text{S-18})$$

S-1.3.3 Effective Langmuir constant

Define $Q_{\max}^{(\text{tot})} = Q_{\max,1} + Q_{\max,2}$. Then

$$K_L^{(\text{tot})} = \frac{K_H^{(\text{tot})}}{Q_{\max}^{(\text{tot})}} = \frac{Q_{\max,1} K_1 + Q_{\max,2} K_2}{Q_{\max,1} + Q_{\max,2}}, \quad (\text{S-19})$$

which is a capacity-weighted average of K_1 and K_2 .

S-1.3.4 Thermodynamic interpretation

With a standard concentration C^0 ,

$$K_i^* = K_i C^0, \quad K_L^{(\text{tot})*} = K_L^{(\text{tot})} C^0. \quad (\text{S-20})$$

116 The effective adsorption free energy and affinity are

$$\Delta G_{\text{ads}}^{0,(\text{tot})} = -RT \ln \left(K_L^{(\text{tot})} C^0 \right), \quad E_{\text{aff}}^{(\text{tot})} = RT \ln \left(K_L^{(\text{tot})} C^0 \right). \quad (\text{S-21})$$

117 **S-1.4 Freundlich isotherm**

118 The Freundlich isotherm is

$$q_e = K_F C_e^{1/n}, \quad (\text{S-22})$$

119 where K_F is a capacity-related constant and $1/n$ is the heterogeneity exponent.

120 **S-1.4.1 High concentration limit: no intrinsic Q_{max}**

121 Taking $C_e \rightarrow \infty$ in Eq. (S-22),

$$\lim_{C_e \rightarrow \infty} q_e = \infty. \quad (\text{S-23})$$

122 Thus, the Freundlich model has no finite saturation capacity:

Warning

Freundlich has no intrinsic Q_{max} . (S-24)

124 **S-1.4.2 Low concentration behavior and Henry law**

125 The derivative of Eq. (S-22) is

$$\frac{dq_e}{dC_e} = K_F \frac{1}{n} C_e^{1/n-1}. \quad (\text{S-25})$$

126 For $1/n = 1$, the Freundlich model reduces to Henry law with $K_H = K_F$. For $1/n \neq 1$,
 127 the initial slope diverges ($1/n < 1$) or vanishes ($1/n > 1$) as $C_e \rightarrow 0$. An apparent,
 128 concentration-dependent Henry coefficient can be defined as

$$K_H(C_e) = \frac{q_e}{C_e} = K_F C_e^{1/n-1}. \quad (\text{S-26})$$

129 **S-1.4.3 Affinity constant and apparent free energy**

130 Using an effective Henry coefficient at a reference concentration C_{ref} , one can define

$$K_{\text{eff}}^* = \left[\frac{1}{n} K_F C_{\text{ref}}^{1/n-1} \right] C^0, \quad (\text{S-27})$$

131 leading to an apparent free energy

$$\Delta G_{\text{ads}}^{0,\text{app}}(C_{\text{ref}}) = -RT \ln \left[\frac{1}{n} K_F C_{\text{ref}}^{1/n-1} C^0 \right]. \quad (\text{S-28})$$

132 This dependence on C_{ref} reflects the lack of a unique free energy for the Freundlich model.

Reporting warning

Freundlich can fit curved data, but it does not define a unique Q_{max} , K_{aff}^* , or E_{ads} without an arbitrary concentration reference. Treat Freundlich affinity values as local or apparent, not as universal descriptors.

133

134 **S-1.5 Sips (Langmuir–Freundlich) isotherm**

135 The Sips model introduces a finite saturation capacity while retaining heterogeneity:

$$q_e = \frac{Q_{\text{max}} K_S C_e^m}{1 + K_S C_e^m}, \quad (\text{S-29})$$

136 where $0 < m \leq 1$. When $m = 1$, the model reduces to Langmuir.

137 **S-1.5.1 High concentration limit: finite Q_{max}**

138 Taking $C_e \rightarrow \infty$ in Eq. (S-29),

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\text{max}}. \quad (\text{S-30})$$

139 **S-1.5.2 Low concentration limit: Freundlich behavior**

140 For $K_S C_e^m \ll 1$,

$$q_e \approx Q_{\max} K_S C_e^m. \quad (\text{S-31})$$

141 This has the Freundlich form with

$$K_F = Q_{\max} K_S, \quad \frac{1}{n} = m. \quad (\text{S-32})$$

142 **S-1.5.3 Initial slope and local Henry coefficient**

143 The low-concentration ratio is

$$\frac{q_e}{C_e} \approx Q_{\max} K_S C_e^{m-1}, \quad (\text{S-33})$$

144 so a local Henry coefficient can be defined as

$$K_H(C_e) = Q_{\max} K_S C_e^{m-1}, \quad (\text{S-34})$$

145 which is constant only when $m = 1$.

146 **S-1.5.4 Effective affinity energy**

147 Define an effective equilibrium constant K_{eq} by

$$\frac{E_0}{RT} = \ln(K_{\text{eq}} C^o). \quad (\text{S-35})$$

148 For Langmuir, $K_{\text{eq}} = K_L$, so

$$\text{Langmuir:} \quad \frac{E_0}{RT} = \ln(K_L C^o). \quad (\text{S-36})$$

149 For Sips, the constant K_S has units $[C]^{-m}$, so an effective equilibrium constant with units
 150 $[C]^{-1}$ is

$$K_{\text{eq}}^{(\text{Sips})} = K_S^{1/m}. \quad (\text{S-37})$$

151 Therefore,

$$\frac{E_0}{RT} = \frac{1}{m} \ln(K_S (C^\circ)^m). \quad (\text{S-38})$$

152 In the limit $m = 1$, Eq. (S-38) reduces to Eq. (S-36).

Sips convention

This work uses $q_e = Q_{\text{max}} K_S C_e^m / (1 + K_S C_e^m)$. Therefore K_S has units $[C]^{-m}$, and the comparable affinity is $K_S^{1/m}$, which has units $[C]^{-1}$. This is the quantity used in K_{aff}^* .

153

154 S-1.6 Tóth isotherm

155 The Tóth isotherm is

$$q_e = \frac{q_{\text{max}} b C_e}{[1 + (b C_e)^t]^{1/t}}, \quad (\text{S-39})$$

156 where q_{max} is the saturation capacity, b is the affinity constant, and $0 < t \leq 1$ is the
 157 heterogeneity parameter. When $t = 1$, the model reduces to Langmuir.

158 S-1.6.1 High concentration limit: finite Q_{max}

159 As $C_e \rightarrow \infty$, $(b C_e)^t \gg 1$, so

$$\lim_{C_e \rightarrow \infty} q_e = \lim_{C_e \rightarrow \infty} \frac{q_{\text{max}} b C_e}{(b C_e)^{t/t}} = q_{\text{max}}. \quad (\text{S-40})$$

Capacity-limit result

Tóth has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = q_{\max}. \quad (\text{S-41})$$

This $q_{\max}$ can be reported as $Q_{\max}$, while the shape parameter t controls how strongly the curve departs from Langmuir.

S-1.6.2 Low concentration limit: Henry constant

For $bC_e \ll 1$, $(bC_e)^t \ll 1$ and $[1 + (bC_e)^t]^{1/t} \approx 1$, so

$$q_e \approx q_{\max} b C_e. \quad (\text{S-42})$$

Note

$$K_H = q_{\max} b. \quad (\text{S-43})$$

S-1.6.3 Affinity constant and adsorption free energy

The affinity constant is $K_{\text{aff}} = b$ (units $[C]^{-1}$). With a standard concentration C^0 ,

$$K_{\text{aff}}^* = b C^0, \quad E_{\text{aff}} = RT \ln(b C^0). \quad (\text{S-44})$$

Note

$$\begin{aligned} \Delta G_{\text{ads}}^0 &= -RT \ln(b C^0), \\ E_{\text{aff}} &= RT \ln(b C^0). \end{aligned} \quad (\text{S-45})$$

167 S-1.7 Jovanovic isotherm

168 The Jovanovic isotherm is

$$q_e = q_{\max} [1 - \exp(-b C_e)]. \quad (\text{S-46})$$

169 S-1.7.1 High concentration limit: finite $Q_{\max}$

170 As $C_e \rightarrow \infty$, $\exp(-b C_e) \rightarrow 0$, so

$$\lim_{C_e \rightarrow \infty} q_e = q_{\max}. \quad (\text{S-47})$$

Capacity-limit result

Jovanovic has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = q_{\max}. \quad (\text{S-48})$$

For reporting, $q_{\max}$ plays the same capacity role as $Q_{\max}$ in Langmuir-type models.

172 S-1.7.2 Low concentration limit: Henry constant

173 For $b C_e \ll 1$, $\exp(-b C_e) \approx 1 - b C_e$, so

$$q_e \approx q_{\max} b C_e. \quad (\text{S-49})$$

174 Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = q_{\max} b. \quad (\text{S-50})$$

176 S-1.7.3 Affinity constant and adsorption free energy

177 The affinity constant is $K_{\text{aff}} = b$. With a standard concentration C^0 ,

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$K_{\text{aff}}^* = b C^0, \quad E_{\text{aff}} = RT \ln(b C^0). \quad (\text{S-51})$$

179 S-1.8 Temkin isotherm

180 The Temkin isotherm is

$$q_e = \frac{RT}{b_T} \ln(a_T C_e). \quad (\text{S-52})$$

181 S-1.8.1 High concentration limit: no intrinsic Q_{max}

182 As $C_e \rightarrow \infty$, $\ln(a_T C_e) \rightarrow \infty$, so

$$\lim_{C_e \rightarrow \infty} q_e = \infty. \quad (\text{S-53})$$

Warning

Temkin has no intrinsic Q_{max} . (S-54)

184 The Temkin model is valid only over an intermediate concentration range where the
185 linear-in- $\ln C_e$ approximation holds.

186 S-1.8.2 Low concentration limit: no finite Henry constant

187 The derivative is

$$\frac{dq_e}{dC_e} = \frac{RT}{b_T C_e}, \quad (\text{S-55})$$

188 which diverges as $C_e \rightarrow 0$. Therefore,

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{Temkin has no finite } K_H. \quad (\text{S-56})$$

S-1.8.3 Effective affinity constant and adsorption free energy

An effective affinity constant is defined from the binding constant a_T (units $[C]^{-1}$):

$$K_{\text{aff}} = a_T, \quad K_{\text{aff}}^* = a_T C^0. \quad (\text{S-57})$$

The adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$E_{\text{aff}} = RT \ln(a_T C^0). \quad (\text{S-58})$$

Note that this is an *effective* energy; the Temkin model describes the mid-coverage regime and does not correspond to a well-defined thermodynamic equilibrium constant.

S-1.9 UNILAN isotherm

The UNILAN isotherm (uniform-energy Langmuir) is

$$q_e = \frac{Q_{\text{max}}}{2s} \ln \left(\frac{1 + b_{\text{max}} C_e}{1 + b_{\text{min}} C_e} \right), \quad (\text{S-59})$$

where $s = \frac{1}{2} \ln(b_{\text{max}}/b_{\text{min}})$ is the half-width of the uniform energy distribution.

199 **S-1.9.1 High concentration limit: finite $Q_{\max}$**

200 As $C_e \rightarrow \infty$, $b_{\max}C_e \gg 1$ and $b_{\min}C_e \gg 1$, so

$$q_e \approx \frac{Q_{\max}}{2s} \ln\left(\frac{b_{\max}}{b_{\min}}\right) = \frac{Q_{\max}}{2s} \cdot 2s = Q_{\max}. \quad (\text{S-60})$$

Capacity-limit result

UNILAN has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-61})$$

Here $Q_{\max}$ is retained as the capacity descriptor, while the energy-band width is carried separately by $b_{\max}/b_{\min}$.

201

202 **S-1.9.2 Low concentration limit: Henry constant**

203 For $b_{\max}C_e \ll 1$ and $b_{\min}C_e \ll 1$, $\ln(1+x) \approx x$, so

$$q_e \approx \frac{Q_{\max}}{2s} (b_{\max} - b_{\min}) C_e. \quad (\text{S-62})$$

204 Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = \frac{Q_{\max}}{2s} (b_{\max} - b_{\min}). \quad (\text{S-63})$$

205

206 **S-1.9.3 Affinity constant and adsorption free energy**

207 The natural affinity constant is the geometric mean of the site constants:

$$K_{\text{aff}} = \sqrt{b_{\text{max}} b_{\text{min}}}, \quad K_{\text{aff}}^* = \sqrt{b_{\text{max}} b_{\text{min}}} C^0. \quad (\text{S-64})$$

208 The adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$E_{\text{aff}} = RT \ln \left(\sqrt{b_{\text{max}} b_{\text{min}}} C^0 \right) = \frac{RT}{2} \ln (b_{\text{max}} b_{\text{min}} (C^0)^2). \quad (\text{S-65})$$

210 **S-1.10 Redlich–Peterson isotherm**

211 The Redlich–Peterson isotherm is

$$q_e = \frac{K_{RP} C_e}{1 + a_{RP} C_e^\beta}, \quad (\text{S-66})$$

212 with $0 < \beta \leq 1$.

213 **S-1.10.1 High concentration limit: conditional Q_{max}**

214 As $C_e \rightarrow \infty$,

$$q_e \approx \frac{K_{RP}}{a_{RP}} C_e^{1-\beta}. \quad (\text{S-67})$$

215 If $\beta = 1$ (Langmuir limit): $Q_{\text{max}} = K_{RP}/a_{RP}$ (finite). If $\beta < 1$: $q_e \rightarrow \infty$ (no finite
216 saturation).

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$Q_{\max} = K_{RP}/a_{RP} \quad (\text{only if } \beta = 1). \quad (\text{S-68})$$

S-1.10.2 Low concentration limit: Henry constant

For $a_{RP}C_e^\beta \ll 1$,

$$q_e \approx K_{RP} C_e. \quad (\text{S-69})$$

Therefore, regardless of β ,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = K_{RP}. \quad (\text{S-70})$$

S-1.10.3 Affinity constant and adsorption free energy

Two effective affinity constants are commonly used:

- Henry regime: $K_{\text{aff}} = K_{RP}/Q_{\max} = a_{RP}$ (when $\beta = 1$).
- High- C regime: $K_{\text{aff}} = (a_{RP})^{1/\beta}$.

For the purpose of cross-model comparison, we adopt the Henry-based definition:

Capacity-limit result

This is the saturation capacity that can be reported as a finite capacity descriptor. If the equation includes a condition, the descriptor is valid only under that condition.

$$K_{\text{aff}}^* = \frac{K_{RP}}{Q_{\text{max}}} C^0 = a_{RP} C^0 \quad (\beta = 1), \quad E_{\text{aff}} = RT \ln(a_{RP} C^0). \quad (\text{S-71})$$

227

228 S-1.11 Koble–Corrigan isotherm

229 The Koble–Corrigan isotherm is

$$q_e = \frac{A_{KC} C_e^{1/n}}{1 + B_{KC} C_e^{1/n}}. \quad (\text{S-72})$$

230 This is algebraically equivalent to the Sips isotherm with the identifications $Q_{\text{max}} = A_{KC}/B_{KC}$,

231 $K_S = B_{KC}$, and $m = 1/n$.

232 S-1.11.1 High concentration limit: finite Q_{max}

233 As $C_e \rightarrow \infty$, $B_{KC} C_e^{1/n} \gg 1$, so

$$\lim_{C_e \rightarrow \infty} q_e = \frac{A_{KC}}{B_{KC}}. \quad (\text{S-73})$$

Capacity-limit result

This is the saturation capacity that can be reported as a finite capacity descriptor. If the equation includes a condition, the descriptor is valid only under that condition.

$$Q_{\text{max}} = A_{KC}/B_{KC}. \quad (\text{S-74})$$

234

235 S-1.11.2 Low concentration limit: conditional Henry constant

236 For $B_{KC} C_e^{1/n} \ll 1$,

$$q_e \approx A_{KC} C_e^{1/n}. \quad (\text{S-75})$$

237 A finite K_H exists only when $1/n = 1$: $K_H = A_{KC}$. For $1/n \neq 1$, the initial slope is
 238 concentration dependent.

239 **S-1.11.3 Affinity constant and adsorption free energy**

240 By analogy with Sips, the effective affinity constant is $K_{\text{aff}} = B_{KC}^n$, so

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$K_{\text{aff}}^* = B_{KC}^n C^0, \quad E_{\text{aff}} = nRT \ln(B_{KC} C^0). \quad (\text{S-76})$$

242 **S-1.12 Radke–Prausnitz isotherm**

243 The Radke–Prausnitz isotherm is

$$q_e = \frac{a_{RP} r_{RP} C_e}{a_{RP} + r_{RP} C_e^{1-p}}, \quad (\text{S-77})$$

244 with $0 < p \leq 1$.

245 **S-1.12.1 High concentration limit: conditional Q_{max}**

246 As $C_e \rightarrow \infty$, for $p < 1$:

$$q_e \approx \frac{a_{RP} r_{RP}}{r_{RP}} C_e^p = a_{RP} C_e^p \rightarrow \infty. \quad (\text{S-78})$$

247 For $p = 1$: $q_e \rightarrow a_{RP} r_{RP} / (a_{RP} + r_{RP}) \cdot C_e$, which also diverges.

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{Radke--Prausnitz has no intrinsic } Q_{\max} \text{ for } p < 1. \quad (\text{S-79})$$

S-1.12.2 Low concentration limit: Henry constant

For small C_e , $r_{RP} C_e^{1-p} \ll a_{RP}$, so

$$q_e \approx r_{RP} C_e. \quad (\text{S-80})$$

Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = r_{RP}. \quad (\text{S-81})$$

S-1.12.3 Effective affinity constant and adsorption free energy

Since no unique $Q_{\max}$ exists, the affinity constant is defined directly from the Henry constant:

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$K_{\text{aff}}^* = r_{RP} C^0, \quad E_{\text{aff}} = RT \ln(r_{RP} C^0). \quad (\text{S-82})$$

This is an *effective* energy valid in the low-concentration regime.

257 **S-1.13 Dubinin–Radushkevich (D-R) and Dubinin–Astakhov (D-**
 258 **A)**

259 The D-R/D-A isotherms are

$$q_e = Q_{\max} \exp \left[- \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}} \right], \quad \varepsilon = RT \ln \left(1 + \frac{C_{\text{ref}}}{C_e} \right), \quad (\text{S-83})$$

260 The D-R model corresponds to $n_{DA} = 2$. Here, C_{ref} is a reference concentration (typically
 261 1 mg L⁻¹ or 1 mol L⁻¹) introduced to maintain dimensional consistency, as adding 1 to a
 262 reciprocal concentration with physical units is mathematically invalid.

Normalization convention

The Polanyi potential is written with an explicit reference concentration, $\varepsilon = RT \ln(1 + C_{\text{ref}}/C_e)$, so the logarithm remains dimensionless. The chosen C_{ref} must be reported in the same concentration units as C_e .

263

264 **S-1.13.1 High concentration limit: finite $Q_{\max}$**

265 As $C_e \rightarrow \infty$, $\varepsilon = RT \ln(1 + C_{\text{ref}}/C_e) \rightarrow 0$, so $\exp[-(\varepsilon/E_a)^{n_{DA}}] \rightarrow 1$ and

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-84})$$

Capacity-limit result

D-R/D-A has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-85})$$

The capacity is well defined, but the affinity scale must still be normalised through the declared C_{ref} and C^0 .

266

267 **S-1.13.2 Low concentration limit: no finite Henry constant**

268 As $C_e \rightarrow 0$, $\varepsilon \rightarrow \infty$, so $\exp[-(\varepsilon/E_a)^{n_{DA}}] \rightarrow 0$ faster than any power of C_e . The derivative

$$\frac{dq_e}{dC_e} = Q_{\max} \frac{n_{DA}}{E_a} \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}-1} \exp \left[- \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}} \right] \cdot \frac{RT C_{\text{ref}}}{C_e(C_e + C_{\text{ref}})} \quad (\text{S-86})$$

269 diverges as $C_e \rightarrow 0$.

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{D-R/D-A has no finite } K_H. \quad (\text{S-87})$$

270

271 **S-1.13.3 Effective affinity constant and adsorption free energy**

272 An effective affinity constant is defined from the half-loading concentration $C_{1/2}$ (the con-
273 centration at which $q_e = Q_{\max}/2$). Setting $q_e = Q_{\max}/2$ gives $\varepsilon_{1/2} = E_a(\ln 2)^{1/n_{DA}}$, and

$$C_{1/2} = \frac{1}{\exp(\varepsilon_{1/2}/RT) - 1}. \quad (\text{S-88})$$

274 The effective affinity constant and energy are

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$K_{\text{aff}} = 1/C_{1/2}, \quad K_{\text{aff}}^* = C^0/C_{1/2}, \quad E_{\text{aff}} = RT \ln(C^0/C_{1/2}). \quad (\text{S-89})$$

275

276 S-1.14 Hill isotherm

277 The Hill isotherm is

$$q_e = Q_{\max} \frac{C_e^{n_H}}{K_D + C_e^{n_H}}, \quad (\text{S-90})$$

278 where n_H is the Hill coefficient and K_D is the dissociation constant. When $n_H = 1$, the
279 model reduces to Langmuir with $K_L = 1/K_D$.

280 S-1.14.1 High concentration limit: finite $Q_{\max}$

281 As $C_e \rightarrow \infty$, $C_e^{n_H} \gg K_D$, so

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-91})$$

Capacity-limit result

Hill has a finite saturation capacity:

$$\lim_{C_e \rightarrow \infty} q_e = Q_{\max}. \quad (\text{S-92})$$

Report $Q_{\max}$ as capacity and interpret n_H separately as the cooperativity or heterogeneity-shape parameter.

282

283 S-1.14.2 Low concentration limit: conditional Henry constant

284 For $C_e^{n_H} \ll K_D$,

$$q_e \approx \frac{Q_{\max}}{K_D} C_e^{n_H}. \quad (\text{S-93})$$

285 A finite K_H exists only when $n_H = 1$:

$$K_H = Q_{\max}/K_D \quad (n_H = 1). \quad (\text{S-94})$$

286 For $n_H \neq 1$, the initial slope is concentration dependent:

$$K_H(C_e) = \frac{Q_{\max}}{K_D} C_e^{n_H-1}. \quad (\text{S-95})$$

287 S-1.14.3 Affinity constant and adsorption free energy

288 The effective affinity constant with units $[C]^{-1}$ is $K_{\text{aff}} = K_D^{-1/n_H}$. Therefore,

$$K_{\text{aff}}^* = K_D^{-1/n_H} C^0, \quad (\text{S-96})$$

289 and the adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0 .

$$E_{\text{aff}} = RT \left(-\frac{1}{n_H} \ln K_D + \ln C^0 \right). \quad (\text{S-97})$$

291 In the limit $n_H = 1$, this reduces to the Langmuir expression $E_{\text{aff}} = RT \ln(K_L C^0)$ with
292 $K_L = 1/K_D$.

293 S-2 Error metrics and information criteria

What this section is for

Error metrics answer a statistical question: “which curve follows the data best?” They do not by themselves prove that a model is physically meaningful. A good workflow uses both: residual statistics first, physical descriptor checks second.

295 S-2.1 Residual error metrics

296 Let the residual at point i be

$$r_i = q_i^{\text{exp}} - q_i^{\text{pred}}(\boldsymbol{\theta}; C_{e,i}), \quad \mathbf{r} = (r_1, \dots, r_n)^\top, \quad (\text{S-98})$$

297 where q_i^{exp} and q_i^{pred} are the experimental and predicted adsorption capacities, n is the
 298 number of data points, p is the number of fitted parameters, and

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n r_i^2 = \arg \min_{\boldsymbol{\theta}} \|\mathbf{r}\|_2^2. \quad (\text{S-99})$$

299 Equation (S-99) is the unweighted non-linear least-squares problem assumed throughout the
 300 main manuscript and implemented in the companion notebooks. If the experimental variance
 301 is known to be strongly non-uniform, a weighted objective $\sum_i w_i r_i^2$ with $w_i \propto 1/\sigma_i^2$ may be
 302 preferable, but all metrics below are given for the unweighted case so that different models
 303 remain directly comparable.

304 For convenience, define the sample mean capacity

$$\bar{q} = \frac{1}{n} \sum_{i=1}^n q_i^{\text{exp}}. \quad (\text{S-100})$$

305 S-2.1.1 Weighted residuals and derived error scales

306 If the experimental uncertainty at each point is known or can be estimated, the weighted
 307 least-squares objective is

$$\text{WSSE} = \sum_{i=1}^n w_i r_i^2, \quad w_i = \frac{1}{\sigma_i^2}, \quad (\text{S-101})$$

308 where σ_i is the standard deviation of the measurement at $C_{e,i}$. In that case the natural
 309 likelihood-based discrepancy is the chi-squared statistic

$$\chi^2 = \sum_{i=1}^n \left(\frac{r_i}{\sigma_i} \right)^2, \quad (\text{S-102})$$

310 with reduced chi-squared

$$\chi_{\text{red}}^2 = \frac{\chi^2}{n - p}. \quad (\text{S-103})$$

311 When all σ_i are equal, χ^2 is proportional to SSE and therefore does not alter model ranking.

312 Two additional scalar error scales are often useful:

$$\text{MSE} = \frac{\text{SSE}}{n}, \quad \text{RMSE} = \sqrt{\frac{\text{SSE}}{n}}, \quad (\text{S-104})$$

313 and the residual variance estimate used in parameter-error propagation is

$$s_r^2 = \frac{\text{SSE}}{n - p}. \quad (\text{S-105})$$

314 Equation (S-105) is the unbiased variance estimator for the residuals after accounting for
315 the p fitted parameters.

316 **S-2.1.2 Absolute error metrics**

317 The sum of squared errors (SSE) is the objective function minimised during non-linear
318 regression:

$$\text{SSE} = \sum_{i=1}^n (q_i^{\text{exp}} - q_i^{\text{pred}})^2. \quad (\text{S-106})$$

319 The coefficient of determination R^2 gives the fraction of variance explained by the model:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}, \quad \text{SST} = \sum_i (q_i^{\text{exp}} - \bar{q})^2. \quad (\text{S-107})$$

320 The adjusted R^2 penalises for the number of fitted parameters, preventing overfitting from
321 being rewarded:

$$R_{\text{adj}}^2 = 1 - \frac{n - 1}{n - p - 1} (1 - R^2). \quad (\text{S-108})$$

322 The sum of absolute errors (EABS) provides a measure less sensitive to outliers than SSE:

$$\text{EABS} = \sum_i |q_i^{\text{exp}} - q_i^{\text{pred}}|. \quad (\text{S-109})$$

323 The mean absolute residual (RESID) normalises EABS by the number of data points, giving
 324 a typical per-point deviation in the same units as q_e :

$$\text{RESID} = \frac{1}{n} \sum_i |q_i^{\text{exp}} - q_i^{\text{pred}}|. \quad (\text{S-110})$$

325 SSE is sensitive to large deviations because the residuals are squared, whereas EABS and
 326 RESID remain in the original q_e units and are less dominated by a small number of outliers.
 327 For non-linear isotherms, R^2 and R_{adj}^2 are descriptive summaries of fit quality; they should
 328 not be used as the sole basis for model selection because they do not encode any physical
 329 penalty for additional parameters.

330 **S-2.1.3 Relative error metrics**

331 The average relative error in percent (ARED) normalises each residual by the experimental
 332 value, giving equal weight to low- and high-concentration data. This metric is particularly
 333 informative when data span several orders of magnitude:

$$\text{ARED} = \frac{100}{n} \sum_i \left| \frac{q_i^{\text{exp}} - q_i^{\text{pred}}}{q_i^{\text{exp}}} \right|. \quad (\text{S-111})$$

334 The median percentage squared error (MPSED) replaces the mean with a median, making
 335 it robust to individual outliers:

$$\text{MPSED} = 100 \times \text{median} \left[\left(\frac{q_i^{\text{exp}} - q_i^{\text{pred}}}{q_i^{\text{exp}}} \right)^2 \right]. \quad (\text{S-112})$$

336 The HYBRID error function combines squared residuals with a $1/q_i^{\text{exp}}$ weight, amplifying
 337 deviations at low concentrations where q_i^{exp} is small. It also corrects for degrees of freedom
 338 $(n - p)$:

$$\text{HYBRID} = \frac{100}{n - p} \sum_i \frac{(q_i^{\text{exp}} - q_i^{\text{pred}})^2}{q_i^{\text{exp}}}. \quad (\text{S-113})$$

339 ARED, MPSED, and HYBRID require $q_i^{\text{exp}} > 0$. This is consistent with adsorption datasets
 340 after removal of zero-capacity points, which otherwise would introduce singular denominators
 341 in the relative-error metrics.

342 **Recommended combination.** No single metric captures all aspects of fit quality. We
 343 recommend reporting at least SSE (or R_{adj}^2) for absolute quality, ARED for relative quality,
 344 HYBRID for low-concentration sensitivity, and AIC_c (Section S-2.2) for model-complexity
 345 penalisation.

346 S-2.1.4 Covariance-matrix errors, confidence intervals, and bands

347 Let the Jacobian of the fitted model be

$$J_{ij} = \left. \frac{\partial q^{\text{pred}}(C_{e,i}; \boldsymbol{\theta})}{\partial \theta_j} \right|_{\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}}, \quad (\text{S-114})$$

348 and denote by $\mathbf{J}$ the $n \times p$ matrix with entries J_{ij} . For unweighted non-linear least squares,
 349 the parameter covariance matrix is approximated by

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \approx s_r^2 (\mathbf{J}^T \mathbf{J})^{-1}, \quad (\text{S-115})$$

350 where s_r^2 is given by Eq. (S-105). In the weighted case the corresponding approximation
 351 becomes

$$\text{Cov}_w(\hat{\boldsymbol{\theta}}) \approx (\mathbf{J}^T \mathbf{W} \mathbf{J})^{-1}, \quad \mathbf{W} = \text{diag}(w_1, \dots, w_n). \quad (\text{S-116})$$

352 The standard error of parameter θ_j is therefore

$$\text{SE}(\hat{\theta}_j) = \sqrt{\left[\text{Cov}(\hat{\boldsymbol{\theta}})\right]_{jj}}, \quad (\text{S-117})$$

353 and a two-sided $(1 - \alpha)$ confidence interval is

$$\hat{\theta}_j \pm t_{1-\alpha/2, n-p} \text{SE}(\hat{\theta}_j), \quad (\text{S-118})$$

354 where $t_{1-\alpha/2, n-p}$ is the Student- t quantile with $n - p$ degrees of freedom.

355 For any predicted uptake at concentration C_e , define the sensitivity vector

$$\mathbf{g}(C_e) = \nabla_{\boldsymbol{\theta}} q^{\text{pred}}(C_e; \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}. \quad (\text{S-119})$$

356 The covariance-based prediction variance is then

$$\text{Var}[\hat{q}(C_e)] \approx \mathbf{g}(C_e)^\top \text{Cov}(\hat{\boldsymbol{\theta}}) \mathbf{g}(C_e), \quad (\text{S-120})$$

357 which gives the approximate pointwise confidence band

$$\hat{q}(C_e) \pm z_{1-\alpha/2} \sqrt{\text{Var}[\hat{q}(C_e)]}, \quad (\text{S-121})$$

358 with $z_{1-\alpha/2}$ the standard normal quantile. This is the covariance-matrix confidence-band
359 construction used in the companion notebooks for visual comparison of fitted curves.

360 **S-2.1.5 Residual diagnostics, bootstrap, and noise sensitivity**

361 Residual metrics alone do not show whether the fitted model is structurally adequate. The
362 residual sequence $\{r_i\}_{i=1}^n$ should therefore be inspected against C_e (or against fitted values)
363 to verify that no systematic curvature remains. For a well-specified model, the residuals
364 should fluctuate around zero without concentration-dependent trends.

365 Parameter and descriptor uncertainty can be propagated by residual bootstrap. Let

$$r_i^c = r_i - \bar{r}, \quad \bar{r} = \frac{1}{n} \sum_{i=1}^n r_i, \quad (\text{S-122})$$

366 and generate bootstrap datasets

$$q_{i,b}^* = q_i^{\text{pred}}(\hat{\boldsymbol{\theta}}; C_{e,i}) + r_{j_b(i)}^c, \quad j_b(i) \in \{1, \dots, n\}, \quad (\text{S-123})$$

367 where the indices $j_b(i)$ are sampled with replacement. Re-fitting the model to each bootstrap
 368 dataset gives a family of parameter vectors $\hat{\boldsymbol{\theta}}^{(b)}$ and therefore a family of descriptors $d^{(b)} \in$
 369 $\{Q_{\text{max}}, E_{\text{ads}}, \sigma_H\}$. The bootstrap standard deviation of any scalar descriptor d is

$$s_d^{\text{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (d^{(b)} - \bar{d})^2}, \quad \bar{d} = \frac{1}{B} \sum_{b=1}^B d^{(b)}. \quad (\text{S-124})$$

370 Percentile intervals from the empirical distribution of $d^{(b)}$ are a practical way to report
 371 confidence intervals on the extracted descriptors. Explicitly, the bootstrap percentile interval
 372 for any scalar quantity x is

$$\text{CI}_{1-\alpha}^{\text{boot}}(x) = [Q_{\alpha/2}(x^{(b)}), Q_{1-\alpha/2}(x^{(b)})], \quad (\text{S-125})$$

373 where $Q_{\beta}(x^{(b)})$ is the empirical β -quantile of the bootstrap sample.

374 Sensitivity to measurement noise can be examined by perturbing the fitted response with
 375 controlled relative noise levels,

$$q_{i,b}^{(\eta)} = q_i^{\text{pred}}(\hat{\boldsymbol{\theta}}; C_{e,i}) (1 + \eta z_{i,b}), \quad z_{i,b} \sim \mathcal{N}(0, 1), \quad (\text{S-126})$$

376 with $\eta = 0.01, 0.05$, or 0.10 for 1%, 5%, and 10% relative noise, respectively. Parameters
 377 or descriptors whose mean shift under 5% noise exceeds their bootstrap standard deviation

should be regarded as weakly constrained by the available data. For a given descriptor d , the relative noise-sensitivity index can be written as

$$\delta_d^{(\eta)} = \frac{|\bar{d}^{(\eta)} - \hat{d}|}{\max(|\hat{d}|, \varepsilon)}, \quad \bar{d}^{(\eta)} = \frac{1}{B_\eta} \sum_{b=1}^{B_\eta} d_b^{(\eta)}, \quad (\text{S-127})$$

where $\hat{d}$ is the descriptor from the original fit and ε is a small numerical safeguard for near-zero quantities. This makes explicit the error-treatment criterion used in the notebook workflow when testing parameter robustness to synthetic perturbations.

S-2.2 Information criteria for model selection

Assuming independent Gaussian residuals with constant variance σ^2 , the log-likelihood of a fitted model is

$$\log L(\hat{\boldsymbol{\theta}}, \hat{\sigma}^2) = -\frac{n}{2} \left[\ln(2\pi) + 1 + \ln\left(\frac{\text{SSE}}{n}\right) \right], \quad (\text{S-128})$$

where the maximum-likelihood variance estimate is $\hat{\sigma}^2 = \text{SSE}/n$. Up to an additive constant common to all models fitted to the same dataset, minimising the negative log-likelihood is equivalent to minimising $n \ln(\text{SSE}/n)$. This is the basis of the AIC, AIC_c, and BIC expressions below.

$$\text{AIC} = n \ln\left(\frac{\text{SSE}}{n}\right) + 2p, \quad (\text{S-129})$$

$$\text{AIC}_c = \text{AIC} + \frac{2p(p+1)}{n-p-1}, \quad (\text{S-130})$$

$$\text{BIC} = n \ln\left(\frac{\text{SSE}}{n}\right) + p \ln n. \quad (\text{S-131})$$

The model with the *lowest* AIC_c (or BIC) is preferred. Interpretation of differences: $\Delta\text{AIC} < 2$: substantial support for both models; $4 < \Delta\text{AIC} < 7$: considerably less support for the higher-AIC model; $\Delta\text{AIC} > 10$: essentially no support. Only models fitted to the

393 same response variable, using the same data points and the same error definition, should be
 394 compared through information criteria.

395 Akaike weights normalise the relative likelihood of each model:

$$w_i = \frac{\exp(-\Delta\text{AIC}_i/2)}{\sum_j \exp(-\Delta\text{AIC}_j/2)}. \quad (\text{S-132})$$

396 These weights satisfy $\sum_i w_i = 1$ and admit a natural model-averaged prediction,

$$\hat{q}_{\text{avg}}(C_e) = \sum_{i=1}^M w_i \hat{q}_i(C_e), \quad (\text{S-133})$$

397 or, analogously, a model-averaged descriptor $d_{\text{avg}} = \sum_i w_i d_i$ when several models remain
 398 within $\Delta\text{AIC}_c < 2$.

399 S-3 Model selection

Practical selection rule

Choose the simplest model that gives acceptable residuals and physically meaningful descriptors. A lower AIC_c is useful, but it should not overrule impossible quantities such as a fake Q_{max} from a non-saturating equation.

400

401 S-3.1 Ranking procedure

- 402 1. Compute AIC_c for all candidate models; rank by ΔAIC .
- 403 2. For nested pairs (Langmuir $\subset$ Sips, Langmuir $\subset$ Double Langmuir, Langmuir $\subset$ Redlich–
 404 Peterson), apply the F-test:

$$F = \frac{(\text{SSE}_1 - \text{SSE}_2)/(p_2 - p_1)}{\text{SSE}_2/(n - p_2)}. \quad (\text{S-134})$$

405 If $F > F_\alpha(p_2 - p_1, n - p_2)$, the more complex model provides a statistically significant

improvement at confidence level α .

3. Inspect residual plots for systematic patterns.

4. Compute Akaike weights (Eq. S-132).

5. If multiple models have $\Delta\text{AIC} < 2$, report all and consider model averaging.

For completeness, the associated upper-tail p -value is

$$p_F = 1 - F_{\nu_1, \nu_2}(F_{\text{obs}}), \quad \nu_1 = p_2 - p_1, \quad \nu_2 = n - p_2, \quad (\text{S-135})$$

where F_{ν_1, ν_2} is the cumulative distribution function of the F distribution. The F-test is meaningful only when the two candidate models are truly nested, fitted to the same dataset, and optimised with the same residual definition.

S-3.2 Energy-distribution overlap

The overlap coefficient between two normalised energy distributions $\hat{f}_A(E)$ and $\hat{f}_B(E)$ quantifies their similarity:

$$\text{OVL} = \int_{-\infty}^{\infty} \min[\hat{f}_A(E), \hat{f}_B(E)] dE. \quad (\text{S-136})$$

For approximately Gaussian distributions:

$$\text{OVL} \approx 2 \Phi\left(-\frac{|E_c^A - E_c^B|}{2\bar{\sigma}}\right), \quad \bar{\sigma} = \sqrt{\frac{\sigma_A^2 + \sigma_B^2}{2}}. \quad (\text{S-137})$$

For arbitrary distributions represented on a numerical energy grid $\{E_g\}_{g=1}^G$, the overlap is evaluated by quadrature as

$$\text{OVL}_{\text{num}} \approx \sum_{g=1}^{G-1} \min[\hat{f}_A(E_g), \hat{f}_B(E_g)] \Delta E_g, \quad \Delta E_g = E_{g+1} - E_g, \quad (\text{S-138})$$

420 after enforcing unit-area normalisation on each distribution. This numerical form is the
 421 appropriate one for asymmetric or non-standard families such as Tóth, Jovanovic, Redlich–
 422 Peterson, or Radke–Prausnitz, for which a closed-form overlap is generally not available.

423 **Interpretation.**

- 424 • OVL > 0.8: models are practically indistinguishable — prefer the simpler one.
- 425 • 0.3 < OVL < 0.8: partial overlap; use AIC/BIC to discriminate.
- 426 • OVL < 0.3: distributions describe different physics.

427 **S-3.3 Specific comparison criteria**

428 **S-3.3.1 Langmuir vs. Sips.**

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Accept Langmuir if } \sigma_H^{(\text{Sips})} = \frac{\pi RT}{\sqrt{3}} \sqrt{\frac{1}{m^2} - 1} \leq \alpha RT, \quad (\text{S-139})$$

430 with $\alpha \approx 1$ as a practical screening value. Smaller values of α are stricter and collapse
 431 toward the Langmuir limit $m \rightarrow 1$, where $\sigma_H \rightarrow 0$.

432 S-3.3.2 Langmuir vs. Double Langmuir.

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Accept Double Langmuir if } \Delta E_{12} = RT \left| \ln \frac{K_1}{K_2} \right| > \beta RT, \quad (\text{S-140})$$

434 with $\beta \approx 2-3$ ($K_1/K_2 > 7.4-20$).

435 S-3.3.3 Sips vs. Double Langmuir.

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Prefer Sips over DL if } \Delta E_{12} < 2 \sigma_H^{(\text{Sips})}. \quad (\text{S-141})$$

437 S-3.4 Decision tree

438 The model-selection decision tree proceeds in three phases: (i) physical screening of the data
 439 shape, (ii) AIC ranking among applicable models, and (iii) nested-model tests with physical
 440 validation. Figure S-1 gives the complete flowchart.

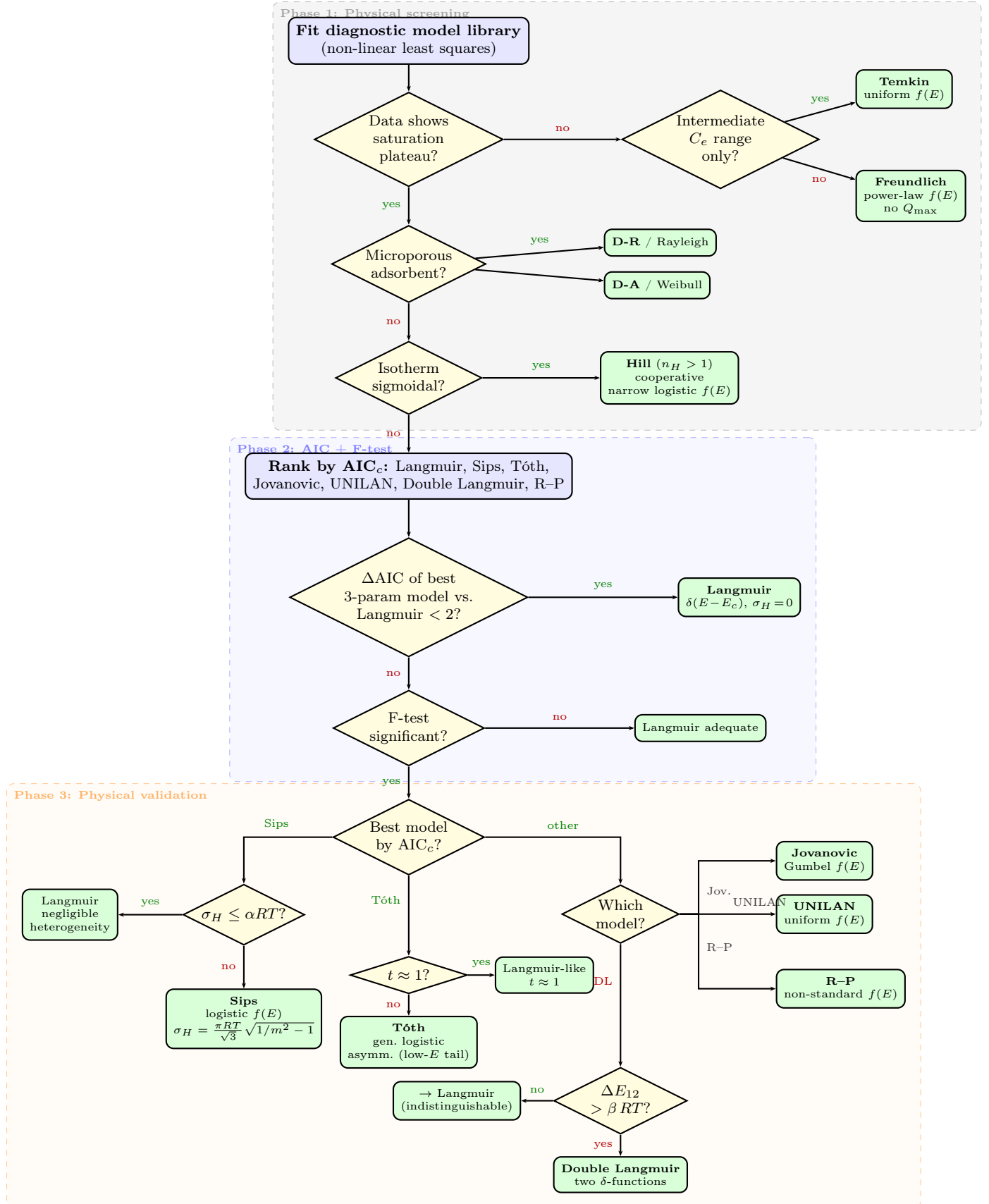


Figure S-1: Complete decision tree for isotherm model selection. **Phase 1** (top): physical screening based on data shape — saturation plateaus, micropore filling, sigmoidal curvature. **Phase 2** (middle): AIC_c ranking and F-test for all saturating models. **Phase 3** (bottom): pairwise tests combined with physical criteria (σ_H , ΔE_{12} , t , β). Green terminal boxes show the accepted model and its energy-distribution family.

441 S-4 Energy distributions of adsorption isotherms

442 S-4.1 General framework

443 A heterogeneous surface is described as a collection of independent Langmuir sites, each with
444 adsorption energy E . The macroscopic isotherm is the integral:

$$q_e = Q_{\max} \int_0^{\infty} \frac{K(E) C_e}{1 + K(E) C_e} f(E) dE, \quad (\text{S-142})$$

445 where $K(E) = K_0 \exp(E/RT)$ and $f(E)$ is the normalised site-energy distribution. Different
446 choices of $f(E)$ yield different macroscopic isotherms.

447 In the *condensation approximation*, the substitution

$$C_e = C^* \exp(-E/RT) \quad (\text{S-143})$$

448 maps the isotherm $q(C_e)$ into $q(E)$, and the energy distribution is recovered by differentiation:

$$f(E) = -\frac{dq}{dE}. \quad (\text{S-144})$$

Width convention

Use σ_E for the apparent width obtained from the mathematical energy distribution. Use σ_H for the intrinsic heterogeneity descriptor reported in the final tables. For logistic-family models (Sips, Koble–Corrigan, Hill), the Langmuir thermal width is

$$\sigma_T = \frac{\pi}{\sqrt{3}} RT, \quad \sigma_H = \sqrt{\max\{\sigma_E^2 - \sigma_T^2, 0\}}.$$

Thus Langmuir and the homogeneous limit of Sips give the same reported value: $\sigma_H = 0$.

449

450 S-4.2 Summary of all energy distributions

451 Table S-1 collects the distribution family, symmetry, and width parameter σ_E for every
452 isotherm model.

Table S-1: Energy-distribution families implied by each isotherm model. All distributions are expressed in the reduced energy $u = (E - E_c)/(RT)$. The standard deviation σ_E is given in units of RT (≈ 2.478 kJ/mol at $T = 298$ K).

Model	Distribution $f(E)$	Symmetry	$\sigma_E/(RT)$	Fit parameters
Langmuir	Dirac $\delta(E - E_c)$	—	0	$K_L, Q_{\max}$
Double Langmuir	Two Dirac δ	—	$\frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} \frac{ \Delta E_{12} }{RT}$	K_1, K_2, Q_1, Q_2
Sips (L-F)	Logistic	Symmetric	$\pi/(m\sqrt{3})$	$K_S, m, Q_{\max}$
Tóth	Gen. logistic (IV)	Asymmetric	$> \pi/\sqrt{3}$ ($t < 1$)	$b, t, q_{\max}$
Jovanovic	Gumbel (EV-I)	Asymmetric	$\pi/\sqrt{6} \approx 1.28$	$b, q_{\max}$
Temkin	Uniform (rect.)	Symmetric	$\frac{b_T Q_{\max}}{2\sqrt{3} RT}$	a_T, b_T
UNILAN	Uniform (rect.)	Symmetric	$\frac{\ln(b_{\max}/b_{\min})}{2\sqrt{3}}$	$b_{\max}, b_{\min}, q_{\max}$
Freundlich	Power law	—	∞	$K_F, 1/n$
Redlich–Peterson	Non-standard	Asymm. ($\beta < 1$)	—	K_{RP}, a_{RP}, β
Koble–Corrigan	$\equiv$ Sips	Symmetric	$\pi n/\sqrt{3}$	$A_{KC}, B_{KC}, 1/n$
Radke–Prausnitz	Non-standard	Asymmetric	—	a_{RP}, r_{RP}, p
D-R	Rayleigh	Asymmetric	$0.463 E_a/RT$	$Q_{\max}, E_a$
D-A	Weibull	Asymmetric	$\frac{E_a}{RT} \sqrt{\Gamma(1 + \frac{2}{n}) - \Gamma(1 + \frac{1}{n})^2}$	$Q_{\max}, E_a, n_{DA}$
Hill	$\equiv$ Sips	Symmetric	$\pi/(\sqrt{3} n_H)$	$Q_{\max}, K_D, n_H$

453 S-4.3 Numerical values of σ_E at $T = 298$ K

454 Table S-2 gives numerical values of σ_E for representative parameter choices at $T = 298$ K
455 ($RT = 2.478$ kJ mol⁻¹).

Table S-2: Numerical values of σ_E at $T = 298$ K ($RT = 2.478$ kJ mol⁻¹) for representative parameter values.

Model	Parameter	$\sigma_E/(RT)$	σ_E (kJ/mol)
Sips	$m = 1.0$	1.81	4.49
Sips	$m = 0.7$	2.59	6.41
Sips	$m = 0.5$	3.63	8.99
Sips	$m = 0.33$	5.49	13.60
Tóth	$t = 0.7$	2.17	5.38
Tóth	$t = 0.5$	2.65	6.56
Tóth	$t = 0.3$	3.51	8.70
Jovanovic	—	1.28	3.18
Temkin/UNILAN	$s = 2 RT$	1.15	2.86
Temkin/UNILAN	$s = 3 RT$	1.73	4.29
Temkin/UNILAN	$s = 4 RT$	2.31	5.72
Hill	$n_H = 0.5$	3.63	8.99
Hill	$n_H = 1.0$	1.81	4.49
Hill	$n_H = 2.0$	0.91	2.24
Hill	$n_H = 4.0$	0.45	1.12
D-R	$E_a = 10$ kJ/mol	1.87	4.63
D-R	$E_a = 20$ kJ/mol	3.74	9.27

How to use this table

The values in Table S-2 are useful for understanding the shape of the mathematical energy distributions. For final material comparison, use the reporting convention in Table S-4; in particular, logistic-family models are reported with σ_H .

S-4.4 Individual energy distributions and σ_E derivations

The condensation approximation (Section S-4.1) is applied to each isotherm by substituting $C_e = C^* e^{-E/RT}$ and computing $f(E) = -(1/Q_{\max}) dq/dE$. The standard deviation σ_E is then obtained from the variance of the resulting distribution family.

461 **S-4.4.1 Langmuir**

462 **Exact result.** The Langmuir isotherm is obtained *exactly* by setting $f(E) = \delta(E - E_c)$ in
 463 Eq. (S-142):

$$Q_{\max} \int_0^\infty \frac{K(E) C_e}{1 + K(E) C_e} \delta(E - E_c) dE = Q_{\max} \frac{K_L C_e}{1 + K_L C_e}, \quad K_L = K_0 e^{E_c/RT}. \quad (\text{S-145})$$

464 No condensation approximation is needed: the Langmuir isotherm is *exact* for a homogeneous
 465 surface ($\sigma_E = 0$).

Characteristic energy and width.

$$E_c = RT \ln(K_L C^*), \quad \sigma_E = 0. \quad (\text{S-146})$$

466 **S-4.4.2 Double Langmuir**

467 **Distribution.** The double-Langmuir isotherm follows from a bimodal discrete distribution:

$$f(E) = \frac{Q_1}{Q_{\max}} \delta(E - E_1) + \frac{Q_2}{Q_{\max}} \delta(E - E_2), \quad Q_{\max} = Q_1 + Q_2, \quad (\text{S-147})$$

468 with $E_i = RT \ln(K_i C^*)$.

469 **Derivation of σ_E .** For a two-point distribution with weights $w_i = Q_i/Q_{\max}$:

$$\langle E \rangle = w_1 E_1 + w_2 E_2, \quad (\text{S-148})$$

$$\sigma_E^2 = w_1 (E_1 - \langle E \rangle)^2 + w_2 (E_2 - \langle E \rangle)^2 = w_1 w_2 (E_1 - E_2)^2, \quad (\text{S-149})$$

470 therefore

$$\sigma_E = \frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} |\Delta E_{12}|, \quad \Delta E_{12} = E_1 - E_2. \quad (\text{S-150})$$

471 For equal sites ($Q_1 = Q_2$): $\sigma_E = |\Delta E_{12}|/2$.

472 S-4.4.3 Sips (Langmuir–Freundlich)

Starting isotherm.

$$q_e = Q_{\max} \frac{K_S C_e^m}{1 + K_S C_e^m}, \quad 0 < m \leq 1. \quad (\text{S-151})$$

473 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$ and defining $E_c = (RT/m) \ln[K_S(C^*)^m] =$
 474 $RT \ln(K_S^{1/m} C^*)$:

$$q(E) = \frac{Q_{\max}}{1 + e^{m(E-E_c)/RT}}. \quad (\text{S-152})$$

Differentiation.

$$\frac{dq}{dE} = -Q_{\max} \frac{(m/RT) e^{m(E-E_c)/RT}}{[1 + e^{m(E-E_c)/RT}]^2}. \quad (\text{S-153})$$

Distribution.

$$f(E) = \frac{Q_{\max} (m/RT) e^{m(E-E_c)/RT}}{[1 + e^{m(E-E_c)/RT}]^2}. \quad (\text{S-154})$$

475 This is the **logistic** probability density (times $Q_{\max}$) with location E_c and scale $s = RT/m$.

476 **Width.** The variance of the logistic distribution with scale s is $\pi^2 s^2/3$, so

$$\sigma_E = \frac{\pi}{\sqrt{3}} \frac{RT}{m}. \quad (\text{S-155})$$

477 For $m = 1$ (Langmuir): $\sigma_E = (\pi/\sqrt{3}) RT \approx 1.81 RT$. As $m \rightarrow 0$: $\sigma_E \rightarrow \infty$ (Freundlich
 478 limit).

479 S-4.4.4 Tóth

Starting isotherm.

$$q_e = \frac{q_{\max} b C_e}{[1 + (b C_e)^t]^{1/t}}, \quad 0 < t \leq 1. \quad (\text{S-156})$$

480 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$, $E_c = RT \ln(bC^*)$, $u = (E -$
 481 $E_c)/RT$:

$$q(E) = q_{\max} \frac{e^{-u}}{[1 + e^{-tu}]^{1/t}}. \quad (\text{S-157})$$

482 **Differentiation.** Applying the product rule and collecting terms:

$$\frac{dq}{dE} = -\frac{q_{\max}}{RT} \frac{e^{-u}}{[1 + e^{-tu}]^{(1+t)/t}}, \quad u = \frac{E - E_c}{RT}. \quad (\text{S-158})$$

Distribution.

$$f(E) = \frac{q_{\max}}{RT} \frac{e^{-u}}{[1 + e^{-tu}]^{(1+t)/t}}, \quad u = \frac{E - E_c}{RT}. \quad (\text{S-159})$$

483 This is the **generalised logistic type IV**, asymmetric with a heavier tail toward *low*
 484 adsorption energies for $t < 1$. For $t = 1$ it reduces to the standard logistic (Langmuir limit).

485 **Width.** No closed-form σ_E exists for general t ; numerical integration is required. $\sigma_E >$
 486 $(\pi/\sqrt{3}) RT$ for all $t < 1$, diverging as $t \rightarrow 0$.

487 **S-4.4.5 Jovanovic**

Starting isotherm.

$$q_e = q_{\max} [1 - \exp(-bC_e)]. \quad (\text{S-160})$$

488 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$, $E_c = RT \ln(bC^*)$, $u = (E -$
 489 $E_c)/RT$:

$$q(E) = q_{\max} [1 - \exp(-e^{-u})]. \quad (\text{S-161})$$

Differentiation.

$$\frac{dq}{dE} = -\frac{q_{\max}}{RT} e^{-u} \exp(-e^{-u}). \quad (\text{S-162})$$

Distribution.

$$f(E) = \frac{q_{\max}}{RT} e^{-u} \exp(-e^{-u}), \quad u = \frac{E - E_c}{RT}. \quad (\text{S-163})$$

490 This is the **Gumbel (extreme-value type I)** distribution with location E_c and scale RT .
 491 The mean is $\langle E \rangle = E_c + \gamma RT$ ($\gamma \approx 0.5772$, Euler–Mascheroni constant). The distribution
 492 is asymmetric with a longer tail toward *high* energies.

493 **Width.** The variance of the Gumbel distribution with scale β is $\pi^2\beta^2/6$:

$$E_c = RT \ln(bC^*), \quad \sigma_E = \frac{\pi}{\sqrt{6}} RT \approx 1.283 RT. \quad (\text{S-164})$$

494 The width is independent of the affinity parameter b and is fixed solely by the model func-
 495 tional form.

496 S-4.4.6 Temkin

Starting isotherm.

$$q_e = \frac{RT}{b_T} \ln(a_T C_e). \quad (\text{S-165})$$

497 (Valid over the intermediate concentration range where the log-linear approximation holds;
 498 implicitly defines $[E_{\min}, E_{\max}]$.)

499 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$:

$$q(E) = \frac{1}{b_T} [RT \ln(a_T C^*) - E]. \quad (\text{S-166})$$

Distribution.

$$f(E) = -\frac{1}{Q_{\max}} \frac{dq}{dE} = \frac{Q_{\max}}{b_T Q_{\max}} = \frac{1}{b_T} = \text{const}, \quad E_{\min} \leq E \leq E_{\max}. \quad (\text{S-167})$$

500 This is a **uniform (rectangular)** distribution over $[E_{\min}, E_{\max}]$, where $E_{\max} = RT \ln(a_T C^*)$
 501 and $E_{\max} - E_{\min} = b_T Q_{\max}$.

502 **Width.** For a uniform distribution on $[a, b]$, $\sigma = (b - a)/(2\sqrt{3})$:

$$\sigma_E = \frac{E_{\max} - E_{\min}}{2\sqrt{3}} = \frac{b_T Q_{\max}}{2\sqrt{3}}. \quad (\text{S-168})$$

503 S-4.4.7 UNILAN

504 **Exact integration.** UNILAN retains the full Langmuir kernel and integrates over a uni-
505 form distribution $f(E) = Q_{\max}/(2s)$ on $[E_c - s, E_c + s]$ analytically:

$$\begin{aligned} q_e &= \frac{Q_{\max}}{2s} \int_{E_c-s}^{E_c+s} \frac{K_0 e^{E/RT} C_e}{1 + K_0 e^{E/RT} C_e} dE \\ &= \frac{Q_{\max} RT}{2s} \ln \frac{1 + b_{\max} C_e}{1 + b_{\min} C_e}, \quad b_{\max/\min} = K_0 e^{(E_c \pm s)/RT}. \end{aligned} \quad (\text{S-169})$$

506 **Distribution and width.** By construction:

$$f(E) = \frac{Q_{\max}}{2s} = \text{const}, \quad s = \frac{RT}{2} \ln \frac{b_{\max}}{b_{\min}}. \quad (\text{S-170})$$

507 The standard deviation of the uniform distribution on $[E_c - s, E_c + s]$ is $s/\sqrt{3}$:

$$\sigma_E = \frac{s}{\sqrt{3}} = \frac{RT}{2\sqrt{3}} \ln \frac{b_{\max}}{b_{\min}}. \quad (\text{S-171})$$

508 S-4.4.8 Freundlich

Starting isotherm.

$$q_e = K_F C_e^{1/n}, \quad 0 < 1/n < 1. \quad (\text{S-172})$$

509 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$:

$$q(E) = K_F (C^*)^{1/n} e^{-E/(nRT)}. \quad (\text{S-173})$$

Distribution.

$$f(E) = \frac{K_F(C^*)^{1/n}}{nRT} e^{-E/(nRT)}. \quad (\text{S-174})$$

510 This one-sided exponential does *not* normalise to a finite area on $[0, \infty)$, reflecting the absence
511 of a finite $Q_{\max}$.

512 **Width.** $\sigma_E = \infty$. The Freundlich model cannot be interpreted as arising from a physically
513 normalised site-energy distribution; it is a purely empirical fitting equation valid only over
514 a limited concentration window.

515 **S-4.4.9 Redlich–Peterson**

Starting isotherm.

$$q_e = \frac{K_{RP} C_e}{1 + a_{RP} C_e^\beta}, \quad 0 < \beta \leq 1. \quad (\text{S-175})$$

516 **Condensation substitution and distribution.** Setting $C_e = C^* e^{-E/RT}$ and differenti-
517 ating:

$$f(E) = \frac{K_{RP} C^* e^{-E/RT}}{Q_{\max} RT} \cdot \frac{1 + (1 - \beta) a_{RP} (C^*)^\beta e^{-\beta E/RT}}{[1 + a_{RP} (C^*)^\beta e^{-\beta E/RT}]^2}. \quad (\text{S-176})$$

518 For $\beta = 1$: reduces to the logistic density (Langmuir). For $\beta < 1$: non-standard asymmetric
519 form with no closed-form σ_E .

520 **S-4.4.10 Koble–Corrigan**

521 **Equivalence to Sips.** The Koble–Corrigan isotherm $q_e = A_{KC} C_e^m / (1 + B_{KC} C_e^m)$ is
522 algebraically identical to Sips with $Q_{\max} = A_{KC}/B_{KC}$ and $K_S = B_{KC}$, same exponent m .
523 The energy distribution and σ_E are therefore those of Section S-4.4.3:

$$\sigma_E = \frac{\pi}{\sqrt{3}} \frac{RT}{m}. \quad (\text{S-177})$$

524 S-4.4.11 Radke–Prausnitz

Starting isotherm.

$$q_e = \frac{a_{RP} r_{RP} C_e}{a_{RP} + r_{RP} C_e^p}, \quad 0 < p \leq 1. \quad (\text{S-178})$$

525 **Distribution.** Applying the condensation substitution yields a non-standard distribution
 526 without a simple closed form. No analytical σ_E exists; numerical integration is required.
 527 For $p \rightarrow 0$ the distribution approaches a logarithmic form with diverging width; for $p = 1$ it
 528 reduces to the Langmuir (logistic) distribution.

529 S-4.4.12 Dubinin–Radushkevich / Dubinin–Astakhov

530 **Polanyi potential.** D-R and D-A operate in the Polanyi adsorption potential $\varepsilon = RT \ln(1 +$
 531 $C_{\text{ref}}/C_e)$ rather than directly in E . Here C_{ref} is a declared reference concentration in the
 532 same units as C_e , ensuring that the logarithm is dimensionless. The isotherm is

$$q_e = Q_{\max} \exp \left[- \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}} \right]. \quad (\text{S-179})$$

533 **Derivation of $f(\varepsilon)$.** Differentiating with respect to ε :

$$f(\varepsilon) = - \frac{1}{Q_{\max}} \frac{dq}{d\varepsilon} = \frac{n_{DA}}{E_a} \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}-1} \exp \left[- \left(\frac{\varepsilon}{E_a} \right)^{n_{DA}} \right]. \quad (\text{S-180})$$

534 This is the **Weibull** probability density with scale E_a and shape n_{DA} .

Mean adsorption potential.

$$\langle \varepsilon \rangle = \frac{E_a}{n_{DA}} \Gamma \left(\frac{1}{n_{DA}} \right). \quad (\text{S-181})$$

535 D-R ($n_{DA} = 2$): $\langle \varepsilon \rangle = E_a \sqrt{\pi}/2$.

536 **Width.** The variance of the Weibull distribution with scale E_a and shape n_{DA} is $E_a^2[\Gamma(1 +$
 537 $2/n_{DA}) - \Gamma(1 + 1/n_{DA})^2]$, therefore

$$\sigma_\varepsilon = E_a \sqrt{\Gamma\left(1 + \frac{2}{n_{DA}}\right) - \left[\Gamma\left(1 + \frac{1}{n_{DA}}\right)\right]^2}. \quad (\text{S-182})$$

538 D-R ($n_{DA} = 2$): $\Gamma(2) = 1$, $\Gamma(3/2) = \sqrt{\pi}/2$, so

$$\sigma_\varepsilon = E_a \sqrt{1 - \pi/4} \approx 0.4633 E_a. \quad (\text{S-183})$$

539 The distribution is right-skewed (longer tail toward high adsorption potentials).

540 **S-4.4.13 Hill**

Starting isotherm.

$$q_e = \frac{Q_{\max} C_e^{n_H}}{K_D + C_e^{n_H}}. \quad (\text{S-184})$$

541 **Condensation substitution.** Setting $C_e = C^* e^{-E/RT}$ and defining $E_c = RT \ln(K_D^{-1/n_H} C^*)$:

$$q(E) = \frac{Q_{\max}}{1 + e^{n_H(E-E_c)/RT}}. \quad (\text{S-185})$$

Differentiation.

$$\frac{dq}{dE} = -Q_{\max} \frac{(n_H/RT) e^{n_H(E-E_c)/RT}}{[1 + e^{n_H(E-E_c)/RT}]^2}. \quad (\text{S-186})$$

Distribution.

$$f(E) = \frac{Q_{\max} (n_H/RT) e^{n_H(E-E_c)/RT}}{[1 + e^{n_H(E-E_c)/RT}]^2}. \quad (\text{S-187})$$

542 Logistic density (times $Q_{\max}$) with location E_c and scale $s = RT/n_H$; identical in form to

543 Sips (Section S-4.4.3) with $m \rightarrow n_H$.

Width.

$$E_c = RT \ln(K_D^{-1/n_H} C^*), \quad \sigma_E = \frac{\pi}{\sqrt{3}} \frac{RT}{n_H}. \quad (\text{S-188})$$

544 **Physical interpretation.**

- 545 • $n_H < 1$ (anti-cooperative): $\sigma_E > (\pi/\sqrt{3}) RT$ — broader than Langmuir.
- 546 • $n_H = 1$ (non-cooperative): $\sigma_E = (\pi/\sqrt{3}) RT \approx 1.81 RT$.
- 547 • $n_H > 1$ (cooperative binding): $\sigma_E < (\pi/\sqrt{3}) RT$ — narrower distribution; the adsorp-
- 548 tion transition is sharper.

549 S-5 Clustering and classification of results

550 Once the descriptor set $\{Q_{\max}, \ln K_{\text{aff}}^*, \sigma_H\}$ has been extracted for many adsorbent–adsorbate
 551 systems, each material is represented by a feature vector

$$\mathbf{x}_i = (Q_{\max,i}, \ln K_{\text{aff},i}^*, \sigma_{H,i})^\top \in \mathbb{R}^d, \quad d = 3. \quad (\text{S-189})$$

552 Stacking the N materials produces the descriptor matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times d}. \quad (\text{S-190})$$

553 All clustering and classification methods below operate on this common descriptor represen-
 554 tation, so the grouping is driven by physically interpretable quantities rather than by the
 555 algebraic form of the original isotherm equation.

Plain-language interpretation

In this three-number space, $Q_{\max}$ answers “how much?”, $\ln K_{\text{aff}}^*$ answers “how strongly?”, and σ_H answers “how heterogeneous?”. These are the quantities the machine-learning methods see; the original isotherm equation is only the route used to estimate them.

556

S-5.1 Feature standardisation

557

$$\tilde{x}_{i,\ell} = \frac{x_{i,\ell} - \bar{x}_\ell}{s_\ell}, \quad s_\ell = \sqrt{\frac{1}{N-1} \sum_i (x_{i,\ell} - \bar{x}_\ell)^2}. \quad (\text{S-191})$$

558 This transformation centers each descriptor to zero mean and rescales it to unit variance, pre-
559 venting $Q_{\max}$, $\ln K_{\text{aff}}^*$, or σ_H from dominating the geometry solely because of their numerical
560 units.

S-5.2 Categorical covariates and one-hot encoding

561

562 If auxiliary categorical information is available, such as material family, synthesis route, or
563 best-fit isotherm class, it can be appended to the continuous descriptor vector by one-hot
564 encoding. Let $\mathbf{z}_i \in \{0, 1\}^q$ denote the binary indicator block for material i . The augmented
565 feature vector is then

$$\mathbf{x}_i^{\text{aug}} = \begin{bmatrix} \tilde{\mathbf{x}}_i \\ \mathbf{z}_i \end{bmatrix} \in \mathbb{R}^{d+q}. \quad (\text{S-192})$$

566 For a single categorical variable with q levels, the indicator block satisfies $\sum_{j=1}^q z_{ij} = 1$. In
567 a linear classifier, the decision surface in the augmented space is therefore a hyperplane in
568 $\mathbb{R}^{d+q}$, exactly as discussed in the main manuscript.

S-5.3 Distance metric

569

$$d(\mathbf{x}_i, \mathbf{x}_j) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|_2 = \sqrt{\sum_{\ell=1}^d \left(\frac{x_{i,\ell} - x_{j,\ell}}{s_\ell} \right)^2}. \quad (\text{S-193})$$

When one-hot variables are included, the same Euclidean formula is applied to $\mathbf{x}_i^{\text{aug}}$, optionally after scaling the binary block consistently across all samples.

S-5.4 k -Nearest Neighbours (KNN)

Given a query material $\mathbf{x}_q$, assign it to the class most common among its k nearest neighbours:

$$\hat{y}(\mathbf{x}_q) = \arg \max_c \sum_{i \in \mathcal{N}_k(\mathbf{x}_q)} \mathbf{1}[y_i = c]. \quad (\text{S-194})$$

Here $\mathcal{N}_k(\mathbf{x}_q)$ is the set of the k nearest training points under Eq. (S-193). A practical starting value is $k \approx \sqrt{N}$, rounded to the nearest odd integer for binary classification.

The choice of k should be validated by leave-one-out cross-validation (LOOCV):

$$\text{Acc}_{\text{LOO}}(k) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_k^{(-i)}(\mathbf{x}_i) = y_i], \quad (\text{S-195})$$

where $\hat{y}_k^{(-i)}$ denotes the KNN prediction obtained after removing sample i from the training set. The value of k that maximises Acc_{LOO} is preferred. Distance-weighted voting may be used as a refinement, but the unweighted majority rule of Eq. (S-194) is sufficient for the exploratory use cases presented in the manuscript.

S-5.5 Support Vector Machines (SVM)

For binary labels $y_i \in \{-1, +1\}$, the soft-margin SVM finds the maximum-margin separating hyperplane by solving

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C_{\text{reg}} \sum_i \xi_i, \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \quad (\text{S-196})$$

584 The decision function can be written in dual form as

$$f(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i \mathcal{K}(\mathbf{x}_i, \mathbf{x}) + b, \quad \hat{y}(\mathbf{x}) = \text{sign}(f(\mathbf{x})), \quad (\text{S-197})$$

585 where only the samples with $\alpha_i > 0$ are support vectors. For a linear kernel, Eq. (S-197)
 586 defines a plane in the continuous descriptor space or, when Eq. (S-192) is used, a hyperplane
 587 in $\mathbb{R}^{d+q}$.

588 For non-linearly separable data, use the RBF kernel $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$.
 589 Multi-class problems can be handled by one-vs-rest or one-vs-one decompositions, both of
 590 which remain compatible with the same descriptor set.

591 S-5.6 Hierarchical clustering (Ward's linkage)

592 Agglomerative clustering builds a dendrogram by iteratively merging the two closest clusters.
 593 For any cluster C , define the within-cluster sum of squares

$$W(C) = \sum_{i \in C} \|\tilde{\mathbf{x}}_i - \bar{\mathbf{x}}_C\|^2, \quad \bar{\mathbf{x}}_C = \frac{1}{|C|} \sum_{i \in C} \tilde{\mathbf{x}}_i. \quad (\text{S-198})$$

594 Ward's linkage chooses the merge that minimises the increase in within-cluster variance:

$$\Delta(A, B) = W(A \cup B) - W(A) - W(B) = \frac{n_A n_B}{n_A + n_B} \|\bar{\mathbf{x}}_A - \bar{\mathbf{x}}_B\|^2. \quad (\text{S-199})$$

595 The dendrogram cut height determines the number of clusters; the “elbow” in the linkage
 596 distance curve provides a natural choice, and cluster stability can be checked by re-running
 597 the procedure on bootstrap resamples of the descriptor matrix.

598 S-5.7 Classifier validation and cluster reproducibility

599 For supervised classifiers, predictive performance should be evaluated by M -fold cross-
600 validation:

$$\text{Acc}_{\text{CV}} = \frac{1}{M} \sum_{m=1}^M \frac{1}{|T_m|} \sum_{i \in T_m} \mathbf{1}[\hat{y}^{(-m)}(\mathbf{x}_i) = y_i], \quad (\text{S-200})$$

601 where T_m is the test subset in fold m and $\hat{y}^{(-m)}$ is the classifier trained on the remaining folds.
602 If no external class labels exist, the cluster labels obtained from Ward's method can be treated
603 as pseudo-labels and their reproducibility quantified by KNN LOOCV (Eq. (S-195)) or SVM
604 cross-validation (Eq. (S-200)). High reproducibility indicates that the clusters reflect a stable
605 geometric structure in descriptor space rather than an arbitrary cut of the dendrogram.

606 S-5.8 Low-dimensional visualisation

607 Principal component analysis (PCA) provides a compact visual summary of the descriptor
608 matrix. For the standardised matrix $\tilde{\mathbf{X}}$, the sample covariance matrix is

$$\mathbf{S} = \frac{1}{N-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad (\text{S-201})$$

609 where the columns of $\mathbf{V}$ are the principal axes and $\mathbf{\Lambda}$ contains the eigenvalues. Projection
610 onto the first r principal components is

$$\mathbf{z}_i = \mathbf{V}_r^\top \tilde{\mathbf{x}}_i, \quad (\text{S-202})$$

611 which is convenient for plotting cluster structure in two dimensions without changing the
612 underlying classifier or clustering definitions.

613 S-5.9 Workflow summary for clustering

- 614 1. For each material, compute $\{Q_{\text{max}}, K_{\text{aff}}^*, \sigma_H\}$.

- 615 2. Standardise all features (Eq. S-191).
- 616 3. If categorical metadata are used, append them through one-hot encoding (Eq. S-192).
- 617 4. Compute the pairwise distance matrix (Eq. S-193).
- 618 5. Apply hierarchical clustering (Ward’s linkage) and cut the dendrogram.
- 619 6. Validate clusters using KNN leave-one-out cross-validation (Eq. S-195) or SVM with
620 cross-validation (Eq. S-200).
- 621 7. Visualise clusters in 2D projections ($Q_{\max}$, $\ln K_{\text{aff}}^*$) or via PCA (Eq. S-202).

Classification versus clustering

KNN and SVM are **classification** tools: they need labels such as high/medium/low performance or material family. Ward’s linkage is **clustering**: it searches for groups when labels are not known. Use the word “classification” for KNN/SVM and “clustering” for Ward.

S-6 Complete reporting protocol

624 For each adsorbent–adsorbate system, the following information should be reported.

Minimum message for readers

A complete result should let another group reproduce the fit and compare the material. Report the raw conditions, the selected model, the fit quality, and the four final descriptors $\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}$.

626 S-6.1 Minimum reporting set

Table S-3: Minimum reporting set for adsorption isotherm analysis.

Category	Quantity	Units	Source
Experimental	T , pH, ionic strength	K, —, M	Measured
	C° (declared)	mg L ⁻¹ or mol L ⁻¹	Convention
	n (data points)	—	Measured
Model fit	Best model name	—	Step 4
	Fitted parameters θ	model-dependent	Step 2
	R^2 , SSE, ARED, AIC _c	—	Step 3
Universal	$Q_{\max}$	mg g ⁻¹	Step 5
	K_{aff}^*	dimensionless	Step 5
	E_{ads}	kJ mol ⁻¹	Step 5
	σ_H	kJ mol ⁻¹	Step 5
Comparison	OVL (if comparing models)	[0, 1]	Step 6
	Cluster label	—	Step 7

627 S-6.2 Extracting descriptors from each model

Name used in the final report

The final manuscript uses E_{ads} . If an intermediate derivation uses E_{aff} , read it as the same positive affinity-energy descriptor:

$$E_{\text{ads}} = E_{\text{aff}} = RT \ln K_{\text{aff}}^*.$$

628

Table S-4: Extraction of operational reporting descriptors from native isotherm parameters. All expressions assume C° is the declared standard-state concentration.

Model	$K_{\text{aff}} ([C]^{-1})$	$E_{\text{ads}}/(RT)$	$\sigma_H/(RT)$
Langmuir	K_L	$\ln(K_L C^\circ)$	0
Double Langmuir	$K_L^{(\text{tot})} = \frac{Q_1 K_1 + Q_2 K_2}{Q_1 + Q_2}$	$\ln(K_L^{(\text{tot})} C^\circ)$	$\frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} \frac{ \Delta E_{12} }{RT}$
Freundlich	$\frac{1}{n} K_F C_{\text{ref}}^{1/n-1}{}^a$	$\ln \left[\frac{1}{n} K_F C_{\text{ref}}^{1/n-1} C^\circ \right]{}^a$	∞
Sips	$K_S^{1/m}$	$\frac{1}{m} \ln(K_S (C^\circ)^m)$	$\frac{\pi}{\sqrt{3}} \sqrt{\frac{1}{m^2} - 1}$
Tóth	b	$\ln(b C^\circ)$	numerical
Jovanovic	b	$\ln(b C^\circ)$	$\frac{\pi}{\sqrt{6}} \approx 1.28$
Temkin	a_T	$\ln(a_T C^\circ)$	Eq. S-168 ^b
UNILAN	$\sqrt{b_{\text{max}} b_{\text{min}}}$	$\frac{1}{2} \ln(b_{\text{max}} b_{\text{min}} (C^\circ)^2)$	$\frac{\ln(b_{\text{max}}/b_{\text{min}})}{2\sqrt{3}}$
Redlich–Peterson	$a_{RP}{}^c$	$\ln(a_{RP} C^\circ){}^c$	—
Koble–Corrigan	B_{KC}^n	$n \ln(B_{KC} C^\circ)$	$\frac{\pi}{\sqrt{3}} \sqrt{n^2 - 1}$
Radke–Prausnitz	$r_{RP}{}^d$	$\ln(r_{RP} C^\circ){}^d$	—
Hill	K_D^{-1/n_H}	$-\frac{1}{n_H} \ln K_D + \ln C^\circ$	$\frac{\pi}{\sqrt{3}} \sqrt{\max\{1/n_H^2 - 1, 0\}}$
D-R/D-A	$1/C_{1/2}$	$\ln(C^\circ/C_{1/2})$	Eq. S-182

^a Apparent, reference-dependent quantity; no intrinsic Q_{max} or universal affinity exists for Freundlich; any Freundlich affinity is local and reference-dependent.

^b Requires a declared effective Temkin window or $Q_{\text{max}}^{\text{eff}}$; not intrinsic to the two-parameter fit.

^c Henry-based cross-model comparison is valid only in the Langmuir limit $\beta = 1$; otherwise Redlich–Peterson has no finite Q_{max} .

^d Effective low-concentration affinity; Radke–Prausnitz has no finite Q_{max} .

For Sips, Koble–Corrigan, and Hill, the table reports the intrinsic descriptor σ_H , not the apparent logistic width σ_E . For other models without a Langmuir thermal-kernel correction, the reported width is the physically relevant model width or is left undefined when no reliable closed form exists.

S-6.3 Key message

Final reporting fingerprint

Regardless of which isotherm model best fits the data, the final report should reduce the result to the same descriptor set:

$$\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}. \quad (\text{S-203})$$

Here $Q_{\max}$ says how much can be adsorbed, K_{aff}^* and E_{ads} place affinity on a declared standard-state scale, and σ_H reports intrinsic heterogeneity. This four-number fingerprint is the bridge between model fitting, cross-study comparison, uncertainty analysis, and descriptor-based classification.

S-7 Companion notebooks and usage

The repository accompanying this manuscript is intended to let the user fit isotherm data, evaluate competing models, extract descriptors, and build classification datasets without rederiving the equations in this SI.

S-7.1 Repository contents

- `01_isotherm_fitting.ipynb`: fits the candidate-model library to raw adsorption data with columns `Ce` and `qe`, then separates converged diagnostic fits from physically admissible reporting models.
- `02_error_metrics_model_selection.ipynb`: computes error metrics, information criteria, F-tests, and the reporting table of descriptors.
- `03_statistical_validation.ipynb`: performs bootstrap, noise-sensitivity, and descriptor-uncertainty propagation.

- 643 • `04_classification_clustering.ipynb`: applies KNN, SVM, and Ward clustering to
644 descriptor tables.
- 645 • `app_isotherm_fitting.py`: Streamlit app for interactive fitting and export of both
646 fit metrics and descriptor tables.
- 647 • `utils_isotherms.py`: shared implementation of model equations, error metrics, de-
648 scriptor extraction, and distribution utilities.

649 S-7.2 Notebook output blocks used in this SI

650 The notebooks are part of the scientific record, but the SI should not display raw notebook
651 exports or raw $\text{\LaTeX}$ setup code. The reader needs to see what was run, what came out,
652 and where the reproducible file is stored. Therefore, the notebook material in this SI is
653 represented as cleaned notebook-style blocks: short input cells, short output cells, exported
654 figures, exported tables, and a link back to the executable notebook in the repository.

Rule for notebook material in the SI

A notebook block must answer three questions: What file or cell was run? What result did it produce? Which manuscript concept does it support? Long imports, package installation commands, plotting boilerplate, and raw $\text{\LaTeX}$ commands belong in the repository or build notes, not in the reader-facing SI.

656 **Notebook block 1: minimal fitting cell.** This is the type of short code block that
657 belongs in the SI. It shows the scientific operation without reproducing the whole notebook.

Notebook input cell: fitting and descriptor extraction

```
In [1]: import pandas as pd
In [2]: from utils_isotherms import fit_all_models, extract_descriptors
In [3]: data = pd.read_csv("sample_isotherm_data.csv")
In [4]: fits = fit_all_models(data["Ce"], data["qe"])
In [5]: descriptors = extract_descriptors(fits, temperature_K=298.15)
```

658

Notebook output cell: reported descriptor table

```
Out [5]: columns = model, Qmax, K_aff_star, E_ads, sigma_H, AICc, BIC
Out [5]: sigma_H is the reported heterogeneity width; legacy sigma_E is not
used as the final descriptor.
```

659

660 **Notebook block 2: figure export.** Figures should appear in the SI as figures, not as
661 screenshots of the notebook interface. The notebook cell should save the figure with a stable
662 filename; the SI should show the exported figure and explain what it demonstrates.

Notebook input cell: export a fit overlay

```
In [6]: ax = plot_fit_overlay(data, fits["Sips"])
In [7]: ax.figure.savefig("SI_notebook01_fit_overlay.pdf",
bbox_inches="tight")
```

663

Notebook output artifact

```
Out [7]: SI_notebook01_fit_overlay.pdf
```

The corresponding SI figure should use the exported file and caption the scientific point: agreement between measured adsorption data and the selected model, residual structure, or comparison among candidate isotherms.

664

665 **Notebook block 3: table export.** Tables should be exported from the notebook after
666 rounding, unit normalization, and descriptor naming have been checked.

Notebook input cell: export a model-selection table

```
In [8]: report = build_model_selection_table(fits)
In [9]: report.to_csv("SI_notebook02_model_selection.csv", index=False)
In [10]: report.to_latex("SI_notebook02_model_selection_table.tex",
index=False)
```

667

Notebook output artifact

```
Out [10]: SI_notebook02_model_selection_table.tex
```

The SI table should contain the final reported columns only: model name, fitted capacity, standardized affinity, adsorption-energy descriptor, intrinsic heterogeneity width σ_H , and model-selection statistics.

668

669 **Notebook block 4: statistical validation.** Bootstrap and noise-sensitivity results should
670 be reduced to intervals and short diagnostic plots. The SI should not include the entire
671 simulation log.

Notebook input cell: bootstrap summary

```
In [11]: boot = bootstrap_fit(data, model="Sips", n_boot=1000)
In [12]: boot_summary = summarize_descriptor_intervals(boot)
In [13]: boot_summary[["E_ads", "sigma_H"]]
```

672

Notebook output cell: uncertainty statement

```
Out [13]: E_ads median and 95 percent interval
Out [13]: sigma_H median and 95 percent interval
```

The text should report whether the descriptor ranking is stable under bootstrap resampling, not merely whether a flexible model can fit the data.

673

674 **Notebook block 5: classification and clustering.** Classification notebooks should show
675 the descriptor columns and the target labels before showing accuracy, confusion matrices,
676 PCA maps, or clusters.

Notebook input cell: descriptor matrix for classification

```
In [14]: X = descriptors[["Qmax", "K_aff_star", "E_ads", "sigma_H"]]  
In [15]: y = descriptors["group"]  
In [16]: scores = cross_validate_svm_knn(X, y)
```

Notebook output cell: classification summary

```
Out [16]: cross-validated KNN and SVM scores  
Out [16]: confusion matrix and PCA descriptor map exported as figure files
```

The SI should interpret whether the separation is driven by capacity, affinity, adsorption energy, or intrinsic heterogeneity, rather than treating the classifier as a black box.

What not to include

Do not include raw `nbconvert` pages, long import cells, package installation logs, hidden execution state, or LaTeX preamble fragments. Those items are useful for building the supplement but they are not scientific content for the reader. The clean SI artifact is the notebook-style block plus the exported figure or table.

Codex–Overleaf working convention

Run and maintain the analysis in Jupyter. Export figures as stable PDF or PNG files and export summary tables as CSV or cleaned table files. In the SI, include only short notebook-style blocks that explain the reproducible action and the output artifact. Keep the complete notebooks in the repository as the executable supplement.

S-7.3 Notebook-to-SI traceability

Code-to-text rule

Every notebook operation is either a direct implementation of an equation in this SI or a diagnostic visualisation of those same quantities. The notebooks should not introduce new physical descriptors beyond $\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}$.

Table S-5: Traceability between active notebook operations and the SI sections where the corresponding concepts are defined.

Notebook or script	Implemented concept	Where it is explained
01_isotherm_fitting.ipynb	Non-linear fitting of the candidate isotherm equations; overlay plots of experimental and fitted curves; extraction of $\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}$.	Model equations in Section S-1; descriptor extraction in Section S-6.1.
02_error_metrics_model_selection.ipynb	SSE, R^2 , adjusted R^2 , EABS, RESID, ARED, MPSED, HYBRID, AIC, AIC_c , BIC, Akaike weights, F-tests, residual plots, parity plots, metric bar/radar summaries, heatmaps, and covariance confidence bands.	Error metrics and confidence bands in Section S-2; model selection in Section S-3.
03_statistical_validation.ipynb	Residual bootstrap, confidence intervals, synthetic-noise sensitivity, and Monte Carlo propagation from fitted parameters to E_{ads} and σ_H .	Bootstrap and noise sensitivity in Section S-2; final descriptors in Section S-6.1.
04_classification_clustering.ipynb	Standardisation, legacy σ_E -to- σ_H column renaming, one-hot encoding of categorical metadata, KNN, SVM, Ward clustering, cross-validation, confusion matrices, and PCA visualisation.	Clustering and classification in Section S-5.
utils_isotherms.py app_isotherm_fitting.py	Shared implementation of the model equations, including Sips with $K_S C_e^m$, Dubinin with $\varepsilon = RT \ln(1 + C_{\text{ref}}/C_e)$, descriptor extraction, and export of metric and descriptor tables.	Sips and Dubinin conventions in Section S-1; width convention in Section S-4; reporting protocol in Section S-6.1.

Diagnostic graphics such as radar charts, heatmaps, parity plots, residual plots, confusion matrices, and PCA maps are interpretive aids. They do not replace the reporting table; they visualise the residual metrics, uncertainty estimates, or descriptor-space classifications defined above.

687 S-7.4 Recommended usage workflow

- 688 1. Prepare a raw isotherm file in `.csv` or Excel format with columns `Ce` and `qe`.
- 689 2. Fit all candidate models using `01_isotherm_fitting.ipynb` or the Streamlit app.
- 690 3. Use `02_error_metrics_model_selection.ipynb` to rank models with SSE, AIC_c ,
691 BIC, Akaike weights, F-tests, and the descriptor set $\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}$.
- 692 4. Use `03_statistical_validation.ipynb` to attach bootstrap intervals and sensitivity
693 tests to the fitted parameters and derived descriptors.
- 694 5. When several materials are available, assemble a descriptor table and analyze it with
695 `04_classification_clustering.ipynb`.

696 S-7.5 Input formats

697 For raw fitting, the minimum required columns are `Ce` and `qe`. For classification and clus-
698 tering, the input table should contain at least two numeric descriptor columns and one
699 target column named `group`; optional categorical columns such as material family or best-fit
700 isotherm type are one-hot encoded automatically in the notebook.

701 S-7.6 Sample files

702 The repository includes `sample_isotherm_data.csv` for raw fitting and three descriptor ta-
703 bles for classification: `sample_descriptors_basic.csv`, `sample_descriptors_isotherm.csv`,
704 and `sample_descriptors_full.csv`. Older notebook variants stored in `backup_notebooks/`
705 are archival only and are not part of the active workflow.

706 A Notebook 01: fitting one isotherm

707 **File:** `01_isotherm_fitting.ipynb`.

Purpose of this notebook

Use this notebook when the starting point is one experimental isotherm: a table of equilibrium concentration C_e and adsorption capacity q_e . Its job is to fit the candidate isotherm equations, compare the fitted curves visually, and extract the universal descriptors from each converged model.

Input file

Place a `.csv`, `.xlsx`, or `.xls` file in the notebook folder. The file must contain at least two columns:

- **Ce**: equilibrium concentration, in one declared unit such as mg L^{-1} or mol L^{-1} ;
- **qe**: adsorbed amount, in one declared unit such as mg g^{-1} .

Column names are treated case-insensitively. The notebook removes missing values and keeps only $C_e > 0$ and $q_e \geq 0$.

How to run it

1. Set `DATA_FILE` to the experimental file name.
2. Run the import and data-loading cells.
3. Run the fitting cell. The notebook fits all implemented models by non-linear least squares; the equations are those listed in Section S-1.
4. Inspect the fit plot and residual plot before accepting the numerical ranking. See Figure S-2
5. Read the descriptor table. The reportable descriptors are $\{Q_{\max}, K_{\text{aff}}^*, E_{\text{ads}}, \sigma_H\}$.

What to look at first

For a materials reader, the first-pass answer is: which model fits without systematic residuals, what $Q_{\max}$ is obtained, whether E_{ads} is comparable to related systems, and whether σ_H is close to zero or clearly heterogeneous.

Common mistake

Do not choose the model only because it gives the highest R^2 . A more flexible model can fit noise. Use this notebook together with `02_error_metrics_model_selection.ipynb`.

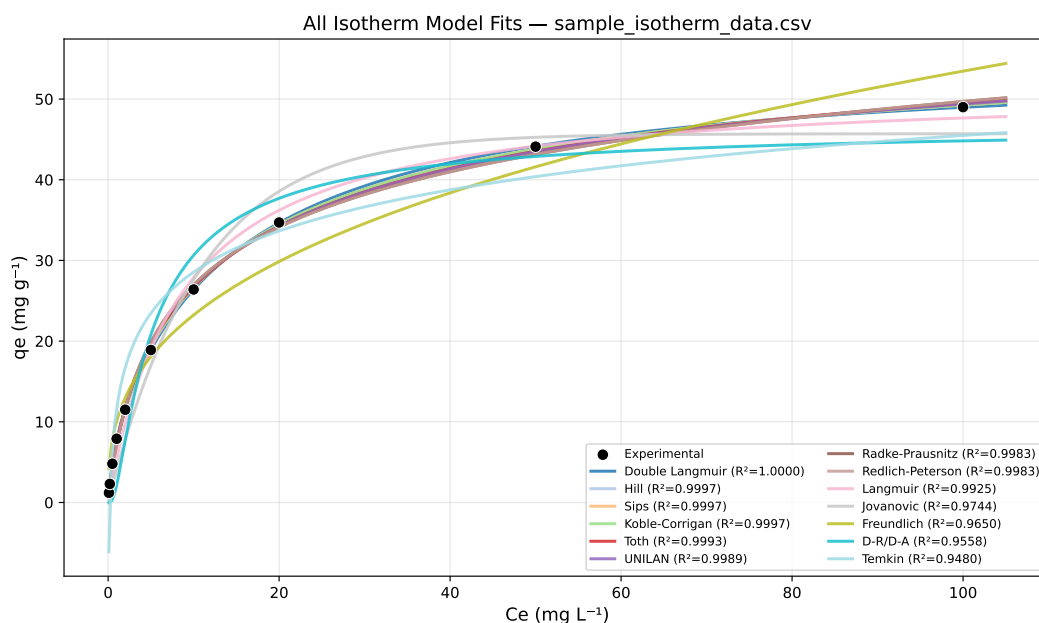


Figure S-2: Overlay of all converged isotherm model fits generated directly from `01_isotherm_fitting.ipynb`. Experimental data are shown together with the fitted model curves, ordered by increasing SSE in the legend.

B Notebook 02: error metrics and model selection

File: `02_error_metrics_model_selection.ipynb`.

Purpose of this notebook

Use this notebook to decide which fitted isotherm should be reported. It computes absolute and relative error metrics, information criteria, Akaike weights, nested-model tests, and diagnostic plots.

Input file

Use the same raw isotherm file as in Appendix A. The same C_e and q_e requirements apply. The notebook refits the models internally so that all metrics are computed from the same dataset and the same candidate set.

How to run it

1. Set `DATA_FILE`.
2. Run all fitting and metric cells.
3. Rank models using AIC_c as the primary statistical criterion.
4. Check whether models within $\Delta AIC_c < 2$ are physically distinguishable by their descriptor values and energy distributions.
5. Use F-tests only for nested model pairs, such as Langmuir versus Sips, and only after confirming that both models converged sensibly.

Output plots

The notebook produces fit overlays, residual plots, grouped metric bars, AIC_c /Akaike-weight summaries, radar charts, parity plots, heatmaps, and covariance-based confidence bands. These figures are diagnostic aids:

- residual plots reveal systematic model failure; See Figure S-3

- parity plots show whether predictions follow the experimental values; See Figure S-4
- heatmaps and radar charts summarize many metrics compactly; See Figure S-5 and Figure S-6
- confidence bands show how much uncertainty is implied by the local covariance matrix. See Figure S-7

Decision rule

Report the simplest physically meaningful model among the statistically supported candidates. A lower AIC_c is useful only if the resulting parameters and descriptors remain physically interpretable.

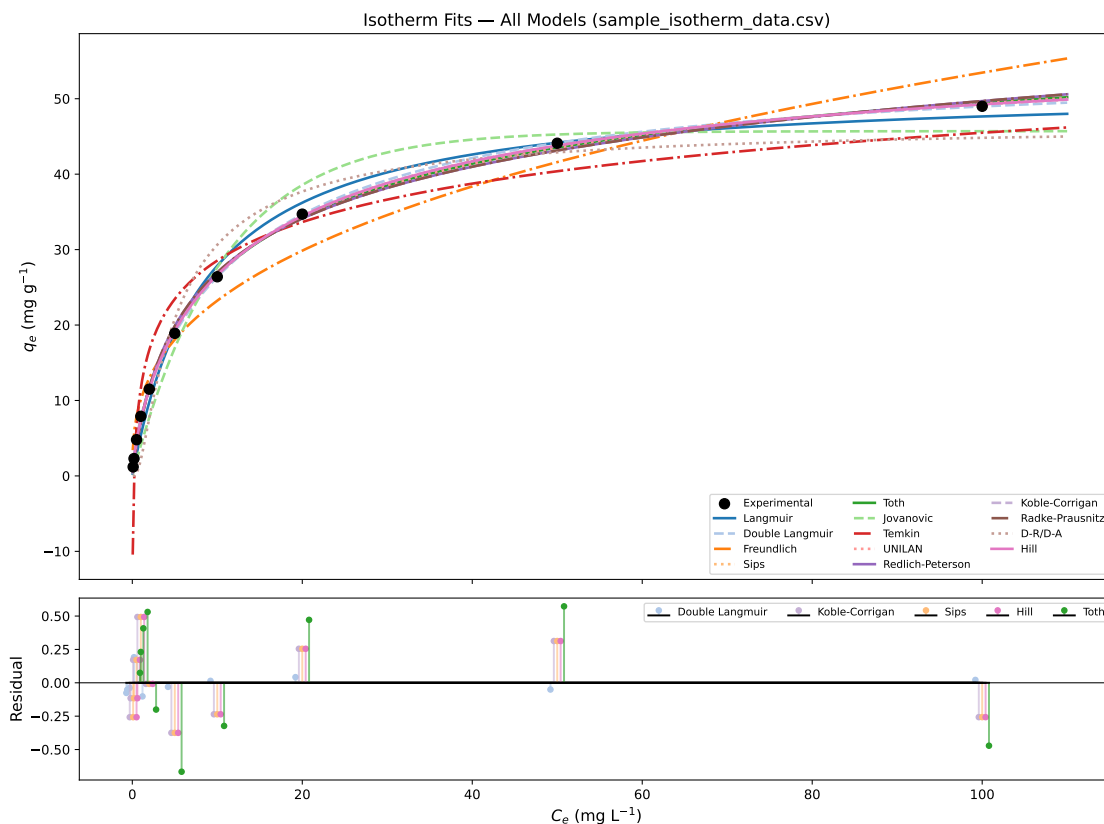


Figure S-3: All converged isotherm model fits and residual analysis generated directly from `01_isotherm_fitting.ipynb`. The upper panel shows the experimental adsorption data together with the fitted model curves. The lower panel shows residuals for the five best-ranked models according to AIC_c .

Residual Diagnostics — Top 4 Models by AICc

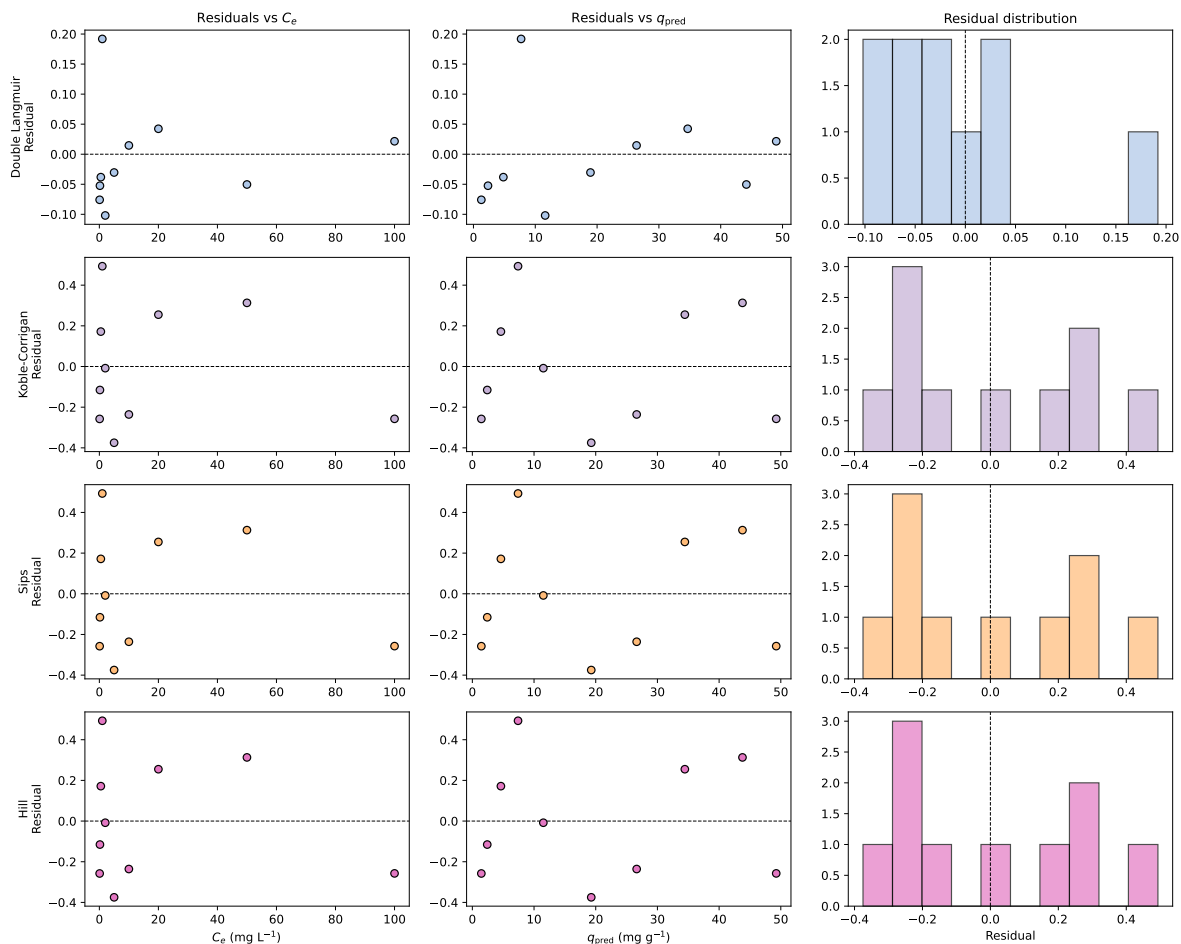


Figure S-4: Residual diagnostics for the four best-ranked isotherm models according to AICc. Each row corresponds to one model. The columns show residuals as a function of equilibrium concentration, residuals as a function of predicted adsorption capacity, and the residual distribution.

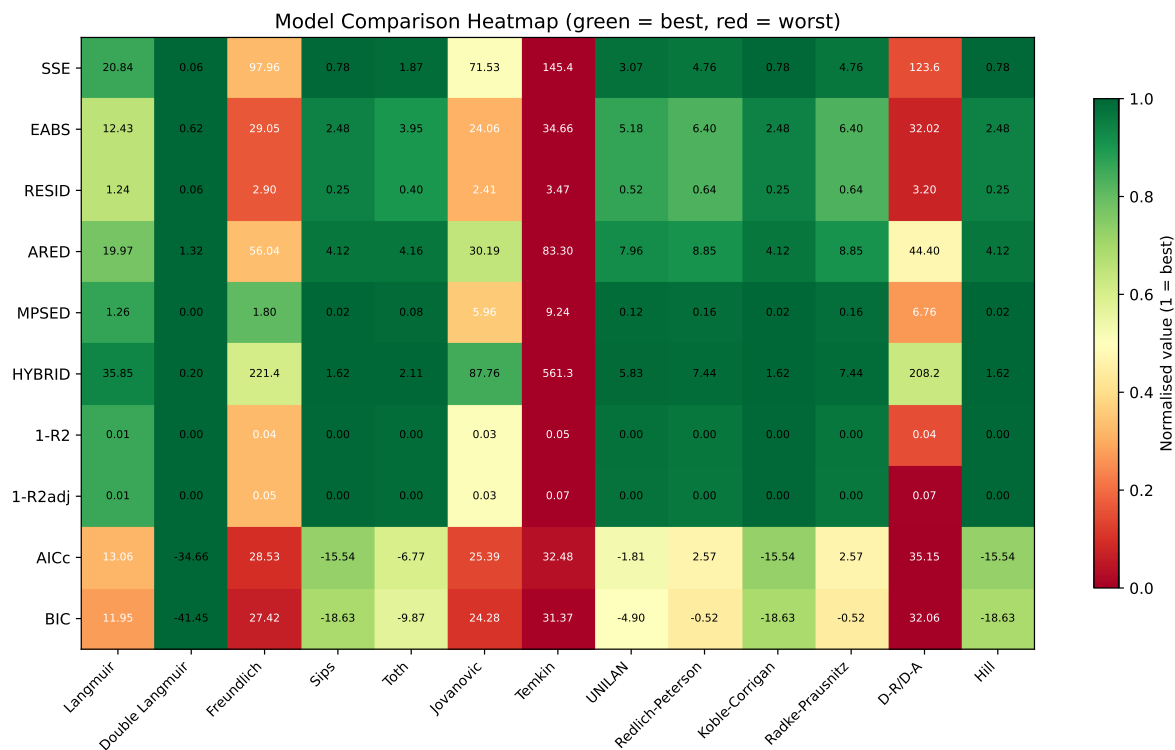


Figure S-5: Model comparison heatmap generated directly from 01_isotherm_fitting.ipynb. Each row corresponds to a fitting metric or information criterion, normalized independently so that green indicates the best-performing model and red indicates the worst-performing model for that metric.

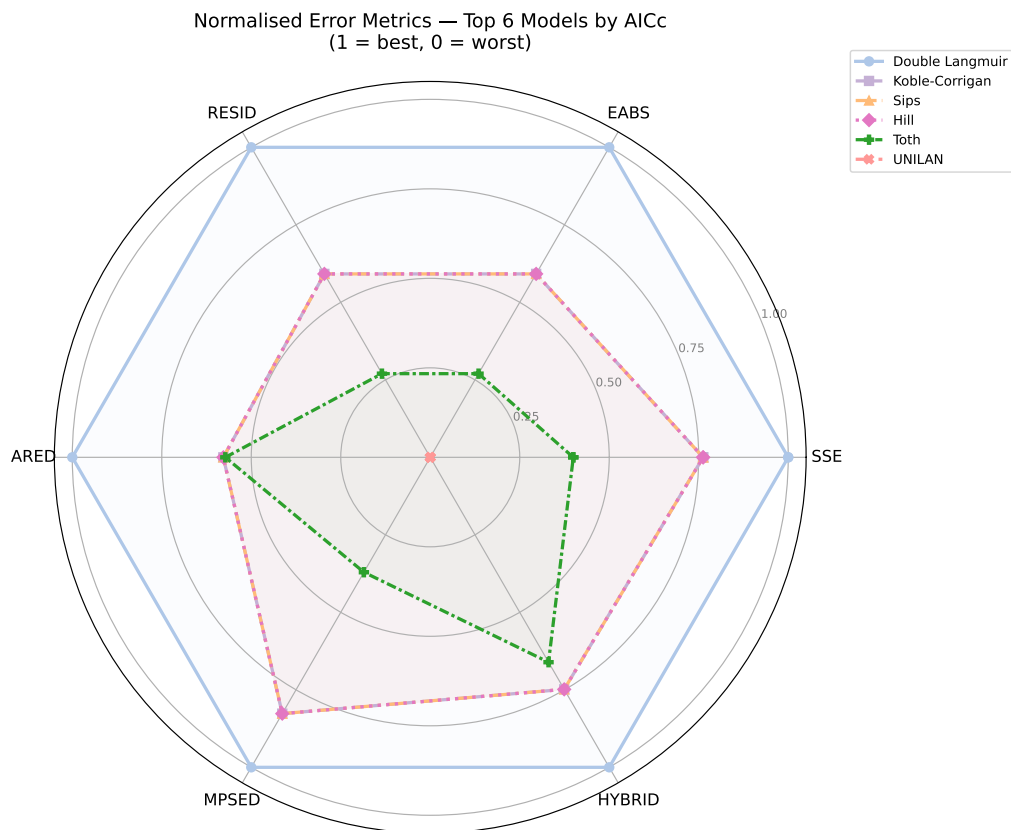


Figure S-6: Radar chart comparing the six best-ranked isotherm models according to AICc. The metrics were independently normalized so that 1 indicates the best-performing model and 0 the worst-performing model for each error criterion.

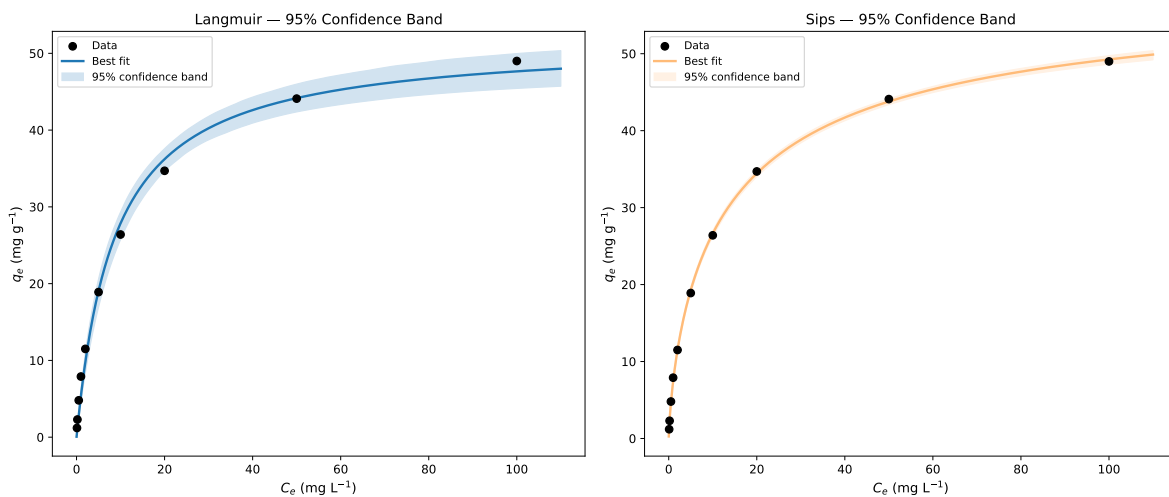


Figure S-7: Confidence bands for Langmuir and Sips isotherm fits generated directly from `01_isotherm_fitting.ipynb`. The solid lines show the best-fit curves, while the shaded regions indicate the 95% confidence bands estimated by Monte Carlo sampling from the fitted parameter covariance matrices.

752 C Notebook 03: statistical validation

753 File: 03_statistical_validation.ipynb.

Purpose of this notebook

Use this notebook after a model has been selected. It asks whether the fitted parameters and descriptors are stable with respect to residual resampling and added measurement noise.

755 What it tests

756 The notebook focuses on the Sips model as a worked example because Sips links directly to
757 the homogeneous Langmuir limit through $m \rightarrow 1$ and $\sigma_H \rightarrow 0$. The same validation logic
758 applies to other fitted models:

- 759 • residual bootstrap gives empirical confidence intervals;
- 760 • synthetic-noise tests show whether the fitted parameters drift when the response is
761 perturbed;
- 762 • Monte Carlo propagation converts parameter uncertainty into uncertainty in E_{ads} and
763 σ_H .

764 How to use the result

- 765 1. Run the baseline fit.
- 766 2. Run the bootstrap cell and check the confidence intervals for Q_{max} , K_S , and m .
- 767 3. Run the noise-sensitivity cell and check whether the parameter shift is small compared
768 with the bootstrap spread.
- 769 4. Report descriptor uncertainty for E_{ads} and σ_H , not only fitted-parameter uncertainty.

Interpretation warning

A narrow confidence interval on a fitted parameter does not automatically mean that the physical descriptor is robust. Always inspect propagated uncertainty in E_{ads} and σ_H .

D Notebook 04: classification and clustering

File: 04_classification_clustering.ipynb.

Purpose of this notebook

Use this notebook when several materials, adsorbates, or operating conditions have already been converted into descriptor rows. It does not fit isotherms. It classifies or clusters descriptor tables.

Input file

The input table should contain:

- at least two numeric descriptor columns, preferably selected from Q_{max} , E_{ads} , $\ln K_{\text{aff}}^*$, and σ_H ;
- one target column named **group** for supervised classification;
- optional categorical columns such as **material_type** or **isotherm_type**.

If an older file contains a column named **sigma_E**, the notebook renames it to **sigma_H** for classification. This is only a backward-compatibility step; new tables should use **sigma_H**.

How to run it

1. Set **DATA_FILE** to one of the descriptor tables.
2. Confirm that **TARGET_COL** = "group" or change it to the desired label column.

- 785 3. Run the preprocessing cells. Numeric descriptors are standardised; categorical columns
786 are one-hot encoded.
- 787 4. Run KNN for local-neighbour classification.
- 788 5. Run SVM for margin-based classification.
- 789 6. Run Ward clustering when no trusted labels are available or when the goal is ex-
790 ploratory grouping.
- 791 7. Use PCA only as a visual summary of high-dimensional descriptor space, not as a new
792 physical model.

Classification versus clustering

793 KNN and SVM require labels such as high/medium/low performance or material family.
Ward clustering does not require labels; it asks whether the descriptor space contains
natural groups.

Common mistake

794 Do not mix raw fitted parameters with universal descriptors in the same classification
table. Use descriptors with consistent units and the same declared C° . Otherwise, the
classifier may separate unit conventions rather than materials.